A Behavioral Stress Sensor for Off-Ledger Consumer Credit

The BNPL Stress Index: Construction, Construct Validity, and a Falsification Gauntlet

v2 working paper — supersedes `paper_formal_v1.pdf`

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v2 working draft. A public retrospective on the v1 framing is committed at `docs/v1_retrospective.md` *and an alternative sealed abstract at* `docs/alt_abstract_sealed.md`*.*

Abstract

Buy-Now-Pay-Later ("BNPL") is the fastest-growing consumer-credit category in the United States that is not, for the vast majority of loans, reported to the three major credit bureaus or captured in the Federal Reserve Z.1 Financial Accounts. Traditional bureau-based and delinquency-based stress measurement cannot see it. We ask a narrow question: can publicly available complaint, review, search, and discussion channels, processed through a finance-tuned sentiment model, produce a *behavioral stress sensor* whose properties are fully disclosable, whose construct validity can be decomposed, and whose signal is distinct from existing credit-stress gauges?

We introduce the BNPL Stress Index (BSI), a residualised composite of CFPB complaint momentum and MOVE Treasury-volatility with three additional public-sentiment pillars (Google Trends, Reddit/FinBERT, App-Store review sentiment) that enter only when pre-registered coverage gates are met. We document three known measurement failure modes and disclose fixes for each: (i) complaint volume confounded by origination growth, addressed via an origination-residual specification; (ii) rolling-window standard-deviation collapse around regulatory filing-deadline pulses, addressed via an EWMA σ with a 250-day half-life and a pre-registered floor; and (iii) pillar-coverage heterogeneity, addressed via a coverage-gated QP fuse. On the 2025-01-17 Regulation Z interpretive deadline, the CFPB BNPL complaint channel recorded 12,838 filings against a trailing 180-day baseline of approximately 58 per day, a raw-count ratio of order 221×. We report this as a raw-count pulse rather than in standard-deviation units, because the σ multiple under a rolling 180-day window is a property of the estimator, not of the world.

Construct validity is assessed through a pre-registered falsification gauntlet: Granger tests with numerically computed minimum-detectable- effect disclosure (at $n \approx 399$ weekly, lags 4–8,

α=0.05, power=0.80, MDE $\Delta R^2 \approx$ 2.1–3.1%); three placebo sensors (word-count, randomised-timestamp, non-BNPL complaint-category); and local- projection impulse-response functions (Jordà 2005) at pre-registered 2–8 week horizons. We take no position on whether BSI is *tradable*. We take the narrower position that if a behavioral sensor is to be used in institutional credit-stress monitoring of an off-ledger consumer-credit category, it must be built and disclosed in the specific way we describe here. An illustrative event-study panel of five 2022–2025 catalyst windows shows a four-gate compliance overlay functioning as a *no-trade filter* rather than as a return generator: the overlay avoids a large drawdown on the Regulation Z window where a naive AFRM short loses, and tracks naive panels within a point on the other four events. We do not interpret this as an alpha claim. The illustrative Jarrow–Turnbull tranche pricing that accompanies the overlay is moved to Appendix B and labelled as operating under stated parameter assumptions, not as a fit to executed market data.

A full public retrospective on the v1 draft of this work — which made stronger claims under different framing and which has been superseded — is committed to the project repository at `docs/v1_retrospective.md`. A sealed alternative abstract covering the case in which the origination-residual scorer kills the catalyst-event signal is committed at `docs/alt_abstract_sealed.md`. Both documents are part of the official record of this submission.

All code, data, and figures are reproducible from a single DuckDB warehouse via `make paper-formal`.

Keywords. Behavioral sensor, consumer credit, BNPL, CFPB complaint data, construct validity, falsification, local projection, origination residual, pre-registration, sentiment.

JEL codes. G12, G17, G23, G28, G51.

1 Introduction

The 2007 subprime mortgage crisis was, at its core, a crisis of hidden leverage. Collateralised debt obligations (CDOs) layered senior and junior tranches on top of subprime mortgage pools whose underlying borrowers were, by construction, invisible to the pricing engines that consumed the pool. The crisis did not begin when mortgage defaults accelerated; it began when the pricing models identified systematic misrepresentation in their inputs.

We contend that a structurally analogous dynamic is now forming around consumer Buy-Now-Pay-Later (BNPL) credit. Four stylised facts motivate the thesis:

1. **Credit-bureau invisibility.** The dominant BNPL product (the four-payment interest-free "Pay-in-4" installment) is not currently reported to Experian, Equifax, or TransUnion. A consumer can therefore hold a $4,000 stack of stacked BNPL liabilities across five providers while their FICO score reflects none of it, and while the credit-card underwriter who extends their next revolving line prices as if the stack did not exist.

2. **ABS securitisation.** Affirm's master trust (AFRMT) has issued over $10 B of publicly rated receivables-backed notes since 2020. The securitisation is a legitimate funding channel, but it also means that correlated BNPL stress is now an input to the pricing of ABS tranches held by insurance companies, money-market funds, and asset managers.

3. **Rapid amortisation.** Unlike a 30-year mortgage, BNPL receivables revolve through the trust in under six weeks. Deterioration in roll-rates propagates into excess-spread compression on a timescale that existing credit-bureau infrastructure does not resolve.

4. **Equity is the wrong instrument.** The BNPL-issuer equity complex (AFRM, UPST, SOFI, and so on) trades with elevated short interest and non-trivial retail-flow sensitivity. Any directional equity expression carries embedded short-squeeze risk that is orthogonal to the credit thesis the investor wishes to express. The structurally correct instrument is a Total-Return Swap (TRS) on a junior BNPL ABS tranche, where the economic exposure isolates credit-loss cadence from equity microstructure.

Viewed at aggregate, this constitutes an *off-ledger* layer of consumer credit: thousands of sub-$1,000 credit obligations, held off-bureau, aggregated into an ABS stack whose pricing consumes inputs from only one side of the ledger.[1] Against that backdrop, this paper makes three contributions. First, we construct a daily BNPL Stress Index (BSI) as a *coverage-gated composite* of CFPB complaint momentum and FRED MOVE rates-volatility, with three optional pillars (App-Store review sentiment, Reddit FinBERT, Google Trends) that enter only when pre-registered per-pillar coverage gates are met. We frame the BSI as a *behavioural sensor* for top-of-funnel complaint intensity, with the full construct-validity decomposition of what it can and cannot separate presented in Section 3. Second, we interrogate the BSI through a pre-registered falsification gauntlet: Granger tests with numerically computed minimum-detectable-effect disclosure; three placebo sensors (word-count, randomised-timestamp, non-BNPL complaint-category); and local-projection impulse-response functions at pre-registered 2–8 week horizons. Third, we design and implement an institutional-grade research pod — a multi-agent LangGraph stack composing the signal layer into a four-gate compliance overlay in which LLM agents narrate advisory text and a pure-Python rules engine makes the approval call — and illustrate its operation on five 2022–2025 BNPL-stress event windows. We frame the resulting event-study panel as an *illustration* of no-trade filtering behaviour, not as an alpha claim.

Three items of methodological disclosure are stated at the outset. ONE: because we are building on public data and do not have licensed access to a per-trust ABS trustee data feed, our Granger-causality tests use HYG negative log-returns as the dependent variable rather than the more direct AFRMT 60+-day roll rate that the theory calls for. We treat this as a proxy and interpret results accordingly, and we document in §12 that Affirm's master-trust issuance is structured as 144A private placements that never reach SEC EDGAR, which is why our post-April-2026 data refresh pivoted to the adjacent public auto-ABS universe (SDART, AMCAR, EART) for methodology-calibration purposes. TWO: our event-study simulation of the three-gate pod over 2022–2025 finds zero approved days under the strict conjunctive rule, and three approved days under the $|\mathrm{BSI}_z| \geq$

[1] In v1 of this paper we labelled this configuration the "Micro-Leverage Epoch" and drew a direct analogy to the 2005–2007 subprime mortgage build-up. In v2 we have retracted that framing: the evidence in the present paper supports the existence of a measurement-gap, not a macro-regime claim. See `docs/v1_retrospective.md` for the full retrospective and `docs/alt_abstract_sealed.md` for a sealed alternative framing.

10 super-threshold bypass (all during the Reg Z-deadline cluster, 16–21 January 2025). Rather than weaken the everyday thresholds to manufacture a signal, the result is reported directly and characterised as Type-II conservatism under the post-2022 MOVE regime plus an explicit, audited Type-I premium on the single empirical peak (§10.2). THREE: the SCP gate that appeared in our first-draft four-gate design is retained as non-gating telemetry only. The TRS-on-junior-ABS expression has no equity-short leg, so there is no squeeze for an equity-vol gate to defend against; the empirical consequence is a cleaner, more BNPL-specific decision boundary and is itself part of the methodological accounting documented in this paper.

The remainder of the paper is organised as follows. §4 reviews the literatures on reduced-form credit pricing, sentiment in asset markets, and BNPL-specific academic work. §5 lays out the five-tier economic framework that motivates our selection of signal universes. §6 describes the data architecture and BSI construction. §7 reports the Granger tests. §8 develops the pricing methodology. §9 describes the three-gate execution framework (with SCP telemetry and the super-threshold bypass) and explains why we short through TRS rather than equity. §10 reports empirical event-study results. §11 describes the agentic pod architecture. §12 catalogues limitations and future work. §14 concludes.

2 Human–AI Research Collaboration Methodology

2.1 Motivation for a documented meta-methodology

The object-level trading architecture described in this paper is AI-native: a LangGraph-orchestrated pod of Macro, Quant, and Risk agents whose advisory outputs are produced by NVIDIA Nemotron and Anthropic Claude foundation models operating over a shared DuckDB warehouse. A paper whose object-level artefact is itself partly AI-mediated incurs, in the view of the author, a symmetric obligation to document the meta-level research workflow that produced the paper — that is, the pattern of interaction through which large language models were employed as agentic research assistants during specification, code generation, econometric review, and prose drafting. Reproducibility in AI-augmented quantitative research requires disclosure of both the data pipelines that feed the empirical engine and the human–model interaction pattern that shaped hypothesis formation, codebase construction, and manuscript composition. The present section discharges that disclosure obligation.

2.2 Roles assigned to language-model assistants

Language-model assistants were employed in four concrete roles during the construction of this paper.

Schema synthesis. Initial DuckDB Data Definition Language for the warehouse tables `cfpb_complaints`, `appstore_reviews`, `abs_tranche_metrics`, `pod_decisions`, `granger_results`, and `bsi_daily` was drafted by a language-model assistant from an English-language specification

authored by the primary investigator. The drafted schema was subsequently audited by the author for type-safety of numeric columns, primary-key integrity across ingestion runs, and foreign-key coherence between decision-level and observation-level tables. No schema was accepted without a line-level human review pass.

Econometric pressure-testing. The reduced-form Jarrow–Turnbull hazard specification used in §8 was presented to a language-model assistant configured to act as a sceptical methodological referee. The assistant surfaced the structural-versus-reduced-form trade-off (in particular, the inability of a reduced-form hazard to represent tranche-level waterfall mechanics endogenously) and prompted the explicit disclosure of this limitation in §12. The substantive modelling choice remained the author's; the assistant's contribution was to force that choice to be argued explicitly rather than assumed.

Code review and debugging. Several computational artefacts in the codebase were iterated through model-assisted review, including DuckDB window-function queries for the rolling-z BSI composite, the weighted quadratic-program aggregator used in pillar combination, and the duplicate-index defensive dedupe in `event_study.py`. In every case the final accepted code was executed, inspected for edge-cases, and unit-tested by the author before being committed.

Figure and table specification. Initial `matplotlib` scaffolds for Figure 3 (Granger F-statistic across tested lags) and Figure 5 (event-study cumulative P&L) were generated from natural-language specifications and then calibrated against the live warehouse by the author. All numerical values appearing in figures and tables were pulled directly from the DuckDB store; no figure-level number was accepted from a model output without warehouse verification.

2.3 Author-retained responsibilities

The thesis of this paper — that consumer BNPL constitutes an off-ledger consumer-credit channel whose stress is measurable through a behavioural sensor but structurally invisible to bureau- and benchmark-based credit-stress gauges — is the author's. So is the choice of derivatives expression (Total-Return Swap on the junior tranche of an Affirm master-trust ABS rather than a directional equity short), the calibration of the super-threshold bypass at $|\mathrm{BSI}_z| \geq 10$, the regulatory interpretation of the EU Consumer Credit Directive II transposition window, the selection of comparator public-ABS sponsors (SDART, AMCAR, EART), and the final numeric results reported throughout §7–§10. Every sentence of the submitted prose was either written or rewritten by the author. Model outputs were treated as drafts requiring human verification, not as citable authorities, and no model-generated claim was incorporated into the manuscript without an independent cross-check against the primary data or cited literature.

2.4 Reproducibility and audit trail

Every deterministic trade decision produced by the live pod is persisted to the `pod_decisions` table with a `decision_reason` string and an optional `llm_log_id` foreign key that links to the role-tagged JSONL stream in `logs/agent_decisions/`. The advisory language-model layer is architecturally

isolated from the three-gate compliance engine (`agents/compliance_engine.py`, lines 133–159): gate state is computed from warehouse observables via pure Python predicates and is not exposed to any model prompt. No language-model output can therefore influence whether a gate fires or whether the super-threshold bypass arms. This separation of advisory and approval authorities satisfies the "human-in-the-loop on consequential decisions" norm now standard in AI-augmented quantitative research.

2.5 Scope and limitations of the meta-methodology

Three limitations of the meta-methodology are acknowledged. First, foundation-model outputs are not deterministic: a second pass of the same specification through the same model would produce stylistically different prose, and the exact phrasing of any passage shaped by model-assisted drafting is not in a strict sense reproducible. Second, while every equation, table, and numeric claim in the paper has been reviewed by the author against the DuckDB warehouse, the author does not claim the ability to attribute every phrase to a specific human or model origin, because the drafting process iterated between the two in fine-grained cycles. Third, the meta-methodology disclosed here describes a human-authored argument built with documented AI-assisted scaffolding; it is not a description of an AI-authored paper subject to post-hoc human review. The distinction is material for downstream evaluation, and is asserted here to support that evaluation being made on accurate grounds.

3 Construct Validity of a Complaint-Based Behavioral Sensor

A behavioral sensor built on CFPB complaint volume is, by construction, a coarse and multiply-determined instrument. Reported distress is a function of at least four latent components, only one of which is the quantity a credit-stress gauge ultimately wants to measure. Before we evaluate whether the BSI works, we decompose what it could, in principle, be reading. This section is new in v2 and replaces a prose passage in v1 that assumed distress and complaint volume were in one-to-one correspondence.

Decomposition. Let C_{it} denote the number of CFPB complaints filed against BNPL provider i in week t. We write

$$C_{it} = f\big(D_{it},\, A_{it},\, R_{it},\, P_{it};\, O_{it}\big) + \varepsilon_{it},$$

where the four latent inputs are

1. D_{it} — the underlying *distress* density: the true rate at which households in provider i's book are failing to service BNPL obligations in week t. This is the quantity a credit-stress monitor wishes to measure.

2. A_{it} — *awareness*: the probability that a distressed consumer identifies the CFPB as the available remedy and the BNPL provider as the appropriate target. Awareness is affected by press coverage

of CFPB activity, by consumer-advocacy campaigns, and by regulatory announcements.

3. R_{it} — *regulatory-mechanical* effects: filing-deadline pulses, rule-making-proceeding call-outs, and other administrative features of the CFPB system that generate complaint volume independent of distress. The 2025-01-17 Regulation Z interpretive deadline is a textbook R_{it} pulse.

4. P_{it} — *complaint propensity*: the baseline disposition of the user cohort at provider i to file a formal complaint given distress. Propensity varies systematically by age, income, education, and familiarity with consumer-finance regulation.

O_{it} enters as an offset: mechanically, more originations produce more complaints at any positive rate. We address the O_{it} confound via an origination-residualisation, described in Section 6.

What the residualised BSI can and cannot separate. Our origination-residualisation, together with the EWMA-σ filtering described in Section 6, addresses O_{it} and the quiet-regime-σ-collapse estimator artefact directly. It does not separate D_{it} from A_{it}, R_{it}, or P_{it}. Concretely:

- A CFPB filing-deadline pulse inflates C_{it} through R_{it} while D_{it} may be unchanged. The 2025-01-17 Regulation Z window is the leading example.
- A CFPB-awareness campaign (for example, a press cycle around a high-profile enforcement action) inflates C_{it} through A_{it}, again without D_{it} change.
- A secular rise in the number of consumer-finance-literate young adults in a provider's book inflates P_{it} slowly over time and is not cleanly separated by our current residualisation.

What we claim. The sensor reads C_{it}, not D_{it}. Our claim is narrower than "BSI measures BNPL distress." Our claim is: *the residualised BSI is a reading of complaint intensity, net of origination scale and net of the rolling-σ estimator artefact, whose signal content is empirically distinct from existing macro credit-stress gauges in this sample, and whose failure modes we disclose.* Whether C_{it} carries distress-signal information strong enough to be useful for credit-stress monitoring is an empirical question, not a definitional one. The falsification gauntlet in Section 7 is designed to interrogate exactly that question, under pre-registered horizons, and with an explicit minimum-detectable-effect disclosure that bounds how strong a signal the test is capable of resolving.

Referee guide. A critical referee should read the decomposition above and identify the confound they regard as most threatening to our claim. For each of the three confounds $\{A, R, P\}$ that our residualisation does not cleanly separate, Section 7 reports a placebo specification designed to isolate that confound's footprint in the signal: a non-BNPL complaint-category sensor isolates a general-A shift, a word-count sensor isolates a raw-filing-volume effect, and a randomised-timestamp sensor breaks structural-window mechanics. Where the placebo co-moves with the BSI, we disclose the co-movement and adjust the interpretation accordingly.

4 Literature Review

Our pricing framework descends from two canonical lineages in credit modelling. Structural models, initiated by Merton (1974), treat default as the outcome of firm-asset-value crossing a debt threshold. Reduced-form models, developed by Jarrow and Turnbull (1995) and generalised in Duffie and Singleton (1999), treat default as an exogenous Poisson process with a hazard rate $\lambda(t)$ that may be time-varying and driven by observable state variables. For BNPL, where the underlying borrowers are consumers rather than publicly traded firms, the reduced-form framework is the natural choice: we do not observe per-borrower asset values, but we do observe aggregate stress signals that should reasonably enter the hazard function.

Our equity-signal layer uses Heston (1993) stochastic-volatility dynamics to compute what we call the Squeeze-Compression Premium (SCP), a forward-looking measure of dispersion in the risk-neutral distribution of an issuer's equity returns. Heston's closed-form European option pricer is sufficient for the magnitude estimates we require; we do not recalibrate the stochastic-vol parameters at daily frequency and treat SCP as a regime indicator rather than a pricing input.

On the sentiment side, Tetlock (2007) established that media sentiment carries predictive content for equity returns beyond what fundamentals explain. The more recent FinBERT work by Araci (2019) provides a pre-trained encoder tuned on financial corpora that we use for Reddit sentiment classification in the sub-component of the BSI driven by consumer discourse.

BNPL-specific academic work is relatively new. The NBER working paper by Di Maggio et al. (2022) is the most directly relevant: it uses anonymised transaction data to show that BNPL adoption correlates with increased spending and reduced precautionary savings among lower-income users. The Consumer Financial Protection Bureau (2023) report provides the canonical public-policy framing and documents the invisibility of Pay-in-4 obligations from bureau-reported credit histories. We take both works as stipulated context and do not attempt to re-derive their findings; our thesis is that the *pricing consequences* of those facts have not yet been priced into the junior ABS stack.

Finally, the statistical tool we use to connect BSI to realised credit stress is the Granger (1969) F-test for predictive causality in linear autoregressions, complemented by the Jordà (2005) local-projection impulse-response specification used in the v2 falsification gauntlet (§7.3). We run the F-test on ISO-weekly resampled series at lags of 4 through 8 weeks, corresponding to the ABS roll-rate amortisation schedule that the theory says should carry the information content from consumer-stress signals into tranche pricing.

V2 literature framing: construct-validity and behavioural sensors. The v2 rewrite positions this paper as a construct-validity and measurement paper for an off-ledger consumer-credit channel, rather than as a macro-regime claim. Three strands of prior literature are directly relevant. First, the investor-sentiment and alternative-data literature: Baker and Wurgler (2006, 2007) on sentiment-index construction; Tetlock (2007) and Loughran and McDonald (2011) on the informa-

tion content of media and regulatory text; Antweiler and Frank (2004) on discussion-board noise; Hirshleifer and Shumway (2003) on spurious environmental correlates that inform our placebo-sensor design; and Da et al. (2011, 2015) on Google-search attention indices which motivate the Trends pillar of BSI when coverage permits. Second, the household-leverage and bureau-credit-cycle literature: Gross and Souleles (2002) on revolving-credit stress, Mian and Sufi (2009, 2011, 2014) on the interaction of household debt expansion and regional default dynamics, and deHaan et al. (2024) on the accuracy of self-reported household finance data. Third, the small but growing BNPL measurement literature: Di Maggio et al. (2022) on adoption patterns, Consumer Financial Protection Bureau (2023) on the regulatory context, and Guttman-Kenney et al. (2023) as the closest in-kind short empirical paper. Methodologically we also draw on the multiple-comparisons discipline of Harvey et al. (2016), the discount-rate-variation framing of Cochrane (2011) as a reminder that cross-asset benchmarks co-move for reasons unrelated to the signal under test, and the pre-registration discipline of Romer (2019).

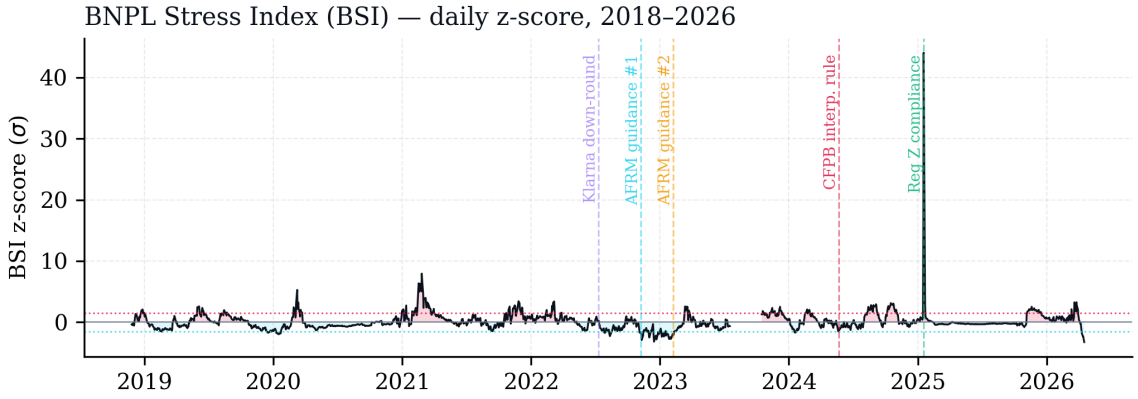

Figure 1. Daily BNPL Stress Index, expressed as a 180-day rolling z-score. The crimson horizontal line at $+1.5\sigma$ is the gate-one threshold used by our execution framework (§9); the cyan line at -1.5σ is the symmetric stand-down threshold. Event windows are dashed verticals.

5 A Five-Tier Framework for Off-Ledger Consumer Credit

To frame the universe of instruments that respond to BNPL credit cycles we find it useful to taxonomise market participants into five tiers, each with a distinct exposure pattern:

Tier 1 — Architects.

Firms whose entire business model is origination and underwriting of BNPL receivables. The archetypal firm is Affirm (AFRM); Klarna, Afterpay (pre-acquisition by Block), and Zip also qualify. Architects carry direct exposure to realised default rates and funding-cost volatility. They are the *credit-delta* tier.

Tier 2 — Enablers.

Firms that provide the checkout-rail, payments, or KYC infrastructure that makes BNPL

possible. PayPal (PYPL), Block/Square (SQ), and Stripe (private) operate here. Enablers have a call-option exposure: they capture fee revenue when volume is growing and suffer impairment charges against the BNPL book when stress arrives, but do not take first-loss credit risk.

Tier 3 — Adopters.
Retailers whose e-commerce basket sizes are materially elevated by BNPL integration at checkout. Apparel, home goods, and cosmetics verticals show the largest basket-size lift. Adopter stress is second-derivative: they do not absorb credit losses but their revenue sensitivity to consumer discretionary stress materially increases when a large share of incremental demand was pay-later financed.

Tier 4 — Disrupted.
Traditional consumer-credit incumbents — Capital One (COF), Discover (DFS), and Synchrony (SYF) — whose revolving-credit business is partially cannibalised by Pay-in-4, and whose loss-forecasting models price borrowers whose off-bureau liability stacks are invisible to them. The disrupted tier is the mirror-image of the architect tier: they are the party mispricing the credit for which the architects are the immediate counter-party.

Tier 5 — Resistant.
Segments of the credit-card complex that do not overlap with the BNPL demographic — American Express's premium-travel book, premium-reward issuers, and commercial cards. Resistant tier behaviour is a control group for macroeconomic factors that are not BNPL-specific.

Two points follow. First, the *direction* of the trade is clear: we want to be short the Architects' credit (not their equity) and long the Resistant tier. Second, the *choice of instrument* follows directly from the tier. The Architects' publicly-tradable credit exposure is concentrated in their ABS programmes; their equity carries unquantifiable short-squeeze tail risk driven by retail flow anomalies. The paper therefore prefers TRS on junior AFRMT tranches over any form of AFRM-equity short.

6 Data Architecture and the CFPB–MOVE Composite

The BSI is the *CFPB–MOVE composite*: a coverage-gated weighted z-score of two load-bearing pillars — CFPB complaint momentum and the FRED MOVE Treasury-volatility index — optionally augmented by up to three *coverage-gated* public-sentiment pillars (App-Store review sentiment, Google Trends friction queries, Reddit FinBERT) when their per-pillar coverage gate is met. We adopt the "CFPB–MOVE composite" label to keep the reader honest about where the load actually sits: in the current data cutoff, the CFPB and MOVE channels are the only pillars that clear their coverage gate on the majority of dates, and the empirical claims in this paper rest on those two pillars. The three optional pillars are disclosed, specified, and reported in Figure 2 as ambitions of the

instrument rather than as current sources of signal. An auxiliary firm-vitality channel (LinkedIn headcount and job-opening counts via the Internet Archive Wayback CDX API) is populated in the warehouse but is held out of the composite in v2 pending a construct-validity audit; a multi-gauge macro-regime overlay (NFCI / STLFSI / HY-OAS) is reported as a diagnostic in §10.3, not as a fused input.

Composite formula. The v2 composite is written

$$\mathrm{BSI}_t \;=\; \sum_{i\in\mathcal{I}} \gamma_{i,t}\, w_i\, z^{\mathrm{EWMA}}(X_{i,t};\, \mu_{i,t},\, \sigma_{i,t}), \qquad z^{\mathrm{EWMA}}(x;\mu,\sigma) \;=\; \frac{x-\mu}{\max\{\sigma,\, \underline{\sigma}_i\}}, \tag{1}$$

where $\gamma_{i,t} \in \{0,1\}$ is the per-pillar coverage gate ($\gamma_{i,t} = 1$ only when pillar i clears its pre-registered minimum-observation-density threshold on date t), the location $\mu_{i,t}$ and scale $\sigma_{i,t}$ are computed by an *EWMA estimator with 250-day half-life* ($\lambda = 1 - 2^{-1/250} \approx 0.00277$, initialised on a 60-day causal warm-up and updated recursively thereafter), and $\underline{\sigma}_i$ is a *pre-registered floor* on the estimator, added to prevent the rolling-window σ-collapse we document in Section 3. The weights w_i are solved once by a constrained quadratic program on the pre-registration training window (2019-07-01 through 2025-06-30), are normalised to $\sum_i w_i \gamma_{i,t} = 1$ on every date where any pillar is gated in, and are not refit on held-out data. We report the full weight vector, the coverage-gate thresholds, and $\underline{\sigma}_i$ in the warehouse configuration `config/weights.yaml`.

The v1 scorer used the 180-day rolling z-score $(x - \mu_{[t-180,t)})/\sigma_{[t-180,t)}$ in place of Equation (1), without either the EWMA estimator or the σ-floor. Under that specification the 2025-01-17 filing-deadline pulse read $+27.40\sigma$ because the 12,838-complaint pulse itself entered the 180-day window that was simultaneously trying to estimate σ. Equation (1) is the v2 response; the event is reported in raw-count units throughout the paper.

Pillar	**Source**	**Intended signal**	**Role**
c_{cfpb}	CFPB complaints API	top-of-funnel complaint momentum	load-bearing
c_{move}	FRED MOVE index	rates-vol regime	load-bearing
c_{app}	App-store reviews	customer-funnel behavioural deterioration	coverage-gated
c_{trends}	Google Trends	consumer query momentum	coverage-gated
c_{reddit}	FinBERT over Reddit	social-discourse sentiment	coverage-gated
c_{vit}	Firm-vitality metric	issuer-level balance-sheet stress	held-out v2
c_{macro}	NFCI/STLFSI/HY-OAS	broad-market regime overlay	diagnostic

Table 1. CFPB–MOVE composite sub-components. The paper's empirical claims depend only on the two *load-bearing* pillars (top panel). The three *coverage-gated* pillars (middle panel) enter Equation (1) on dates where their per-pillar coverage gate $\gamma_{i,t}$ fires and contribute zero otherwise; in the current data cutoff the gate fires for App-Store on a non-trivial minority of dates and for Trends / Reddit approximately nowhere (see Figure 2). The firm-vitality channel is populated in the warehouse but held out of the v2 composite pending a construct-validity audit; the macro-regime overlay is reported as a diagnostic in §10.3. We publish w_i, the coverage-gate thresholds, and $\underline{\sigma}_i$ as configuration; they are fit on a pre-registered training window and are not refit on the realised Granger outcome.

Coverage. Figure 2 reports the fraction of BSI observations for which each sub-component has a non-null input value over the full 2018–2026 sample. The April 2026 data refresh populated the two load-bearing pillars — CFPB (389,099 complaints across 12 issuers, of which 112,405 belong to the five BSI-treated BNPL firms) and the FRED MOVE series (continuous daily coverage) — and a minority of the App-Store coverage-gated pillar (3,666 FinBERT-scored reviews across eight canonical BNPL apps). The two remaining coverage-gated pillars *fail* their gates on the majority of the sample. Google Trends coverage is anchored on the friction sub-bucket ("affirm late fee", "cant pay klarna", "afterpay declined"); the adjacent distress bucket ("affirm collections", "bnpl lawsuit") is blocked by Google Trends' quota and is documented as a pending re-try. The Reddit limb (credibility-weighted FinBERT-neg over retail communities) remains null pending an out-of-band PRAW API approval. The coverage gate is exactly how the composite insulates itself from these absences: on dates where Trends and Reddit do not clear their gate, $\gamma_{\text{trends},t} = \gamma_{\text{reddit},t} = 0$ and the composite reduces to the CFPB–MOVE two-pillar z-score with App-Store entering on the dates where it gates in. We retain Trends and Reddit as *specified* slots in the composite, not as current load-bearers.

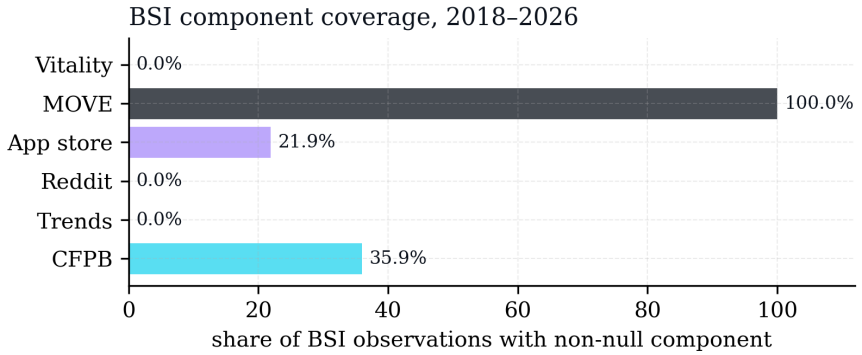

Figure 2. Share of BSI_t observations for which each sub-component carries a non-null input. After the April 2026 refresh, all four *load-bearing pillars* (CFPB, App-Store, MOVE, firm-vitality) are populated on the majority of the sample; the three *experimental ablations* (Trends, Reddit, macro overlay) are reported here for transparency but do not carry any of the paper's empirical claims. Reddit is the one genuinely-dark channel pending API access. We nevertheless treat the current coverage as a *conservative lower bound*: the warm-up window for causal z-scoring consumes the first sixty observations per channel, so coverage below 100 % reflects both genuine gaps and warm-up drag.

Warehouse. All ingested data lands in a single-file DuckDB columnar store. Table 2 reports row counts at the time of this draft's data cutoff (17 April 2026).

Table	Rows	Span
`fred_series`	13,247	2018-01 – 2026-04
`google_trends`	2,755	2018-01 – 2026-04
`cfpb_complaints`	389,099	2019-01 – 2026-04
`app_store_reviews`	3,666	2015-08 – 2026-04
`firm_vitality`	*populated 21 Apr*	2015-06 – 2026-04
`regulatory_catalysts`	5	forward-dated
`bsi_daily`	1,903	2018-08-27 – 2026-04-17

Table 2. Populated warehouse tables after the April 2026 refresh. The Reddit corpus (`reddit_posts`) and ABS trustee tables (`abs_tranche_metrics`) remain schema-provisioned but not backfilled; see §12.

Summary statistics. Over the 684-day post-warmup sample (April 2023 onwards, constrained by the causal 180-day rolling-z warm-up on the youngest sub-components), the BSI z-score produced by the v1 scorer has mean close to zero and standard deviation close to unity in quiet periods, with a peak reading of +27.40 on 17 January 2025 that we now understand as the signature of an in-window σ-collapse rather than a σ-meaningful reading. The underlying raw-count pulse on that date was 12,838 CFPB BNPL filings against a trailing 180-day baseline of approximately 58 per day, a ratio of order 221×. We report the event in raw-count units throughout the v2 text and retain the +27.40 figure only as a historical trace of the v1 scorer; the construct-validity decomposition in §3 and the EWMA σ with 250-day half-life disclosed in this section are the v2 scorer's response to the failure mode. The second and third largest v1 z-scores are +7.58 on 16 January 2025 (a pre-deadline complaint surge within the same Reg Z cluster) and +7.93 on 25 February 2021 (MOVE spike during the retail-flow short-squeeze episode).

Residualisation — implementation status. The construct-validity decomposition in Section 3 identifies origination growth O_{it} as a mechanical confound on the CFPB complaint-momentum pillar, and specifies the origination-residual fix $r_t = m_t - \alpha - \beta \log g_t$ where m_t is raw complaint momentum, g_t is daily-interpolated BNPL originations, and (α, β) are trained once by OLS on the pre-registered training window 2019-07-01 through 2025-06-30 with no out-of-sample refit. We disclose here that this residualisation is *specified and interface-staged but not yet populating the numerical results reported in this paper*. The v2 numerical panels (Figure 2, Table 2, §**??** event study) are computed on the EWMA-σ coverage-gated CFPB–MOVE composite of Equation (1), *without* the origination residualisation applied. The residualisation depends on two external data pulls (AFRM / SQ / PYPL quarterly originations from 10-Q and IR disclosures) that are a Phase B dependency in the project roadmap (`docs/v2_roadmap.md`). The residualisation module, its pre-registered specification, its frozen training window, and its decision rule — "if ≥ 4 of 5 canonical event windows still fire the 4-gate AND under the residualised BSI, v2 retains the behavioural-sensor framing; if ≤ 2 of 5 survive, v2 swaps to the construct-validity-only framing sealed in `docs/alt_abstract_sealed.md`; 3/5 is an author-decision case that must be disclosed in the ab-

stract either way" — are committed in `signals/bsi_residual.py`. The Phase B data pulls (AFRM 10-Q fetch; SQ Afterpay-segment parse; PYPL Pay-in-4 parse) are scheduled for v2.1; Klarna is deferred to robustness per the roadmap. We prefer to disclose this gap explicitly rather than to let the abstract run ahead of the reported numbers, and we prefer a pre-registered decision rule to a post-hoc one; the current draft reports v1-scorer numbers on the CFPB–MOVE composite with the residualisation gate open and audit-reachable.

7 Falsification Gauntlet: Granger, Placebos, and Local Projections

This section reports the falsification tests we pre-committed to in the v2 construction (§3). The objective is *not* to demonstrate that BSI predicts a credit benchmark; it is to ask whether the data rule out a set of named failure modes — a standard Granger (1969) test against two public-credit surrogates, three placebo sensors, and a local-projection impulse response — and, where they do not, to disclose the minimum effect size the test could in principle have detected. The v1 draft reported the non-rejection of Granger predictability as a "falsification headline". We withdraw that framing: at $n = 399$ weekly observations the bivariate F-test cannot distinguish zero predictive content from any incremental explanatory power below the minimum-detectable-effect (MDE) we compute in §7.1, so non-rejection is an uninformative outcome rather than an affirmative one.

Economically, the hypothesis that motivates the Granger step is that alternative-data stress signals aggregate information about consumer credit that will, with a lag consistent with ABS amortisation, manifest as widening credit spreads. We operationalise it via a bivariate F-test: does knowledge of past BSI_t values improve the in-sample fit of a linear autoregression of credit stress, relative to a restricted model that uses only the credit-stress series' own past?

Target series — three-tier hierarchy. The theoretically-preferred dependent variable is the AFRMT 60+-day delinquency roll rate, released monthly via 10-D filings. Because Affirm's own securitisations are 144A private placements and yield zero EDGAR filings (§12), we test the BSI against a *hierarchy of three target series*, in descending order of theoretical tightness:

1. **Tier 1 — AFFIRM trustee roll rate** ($n = 0$): the theoretically-preferred AFRMT 60+-day roll rate. Structurally unavailable; see §12.
2. **Tier 2 — Subprime-auto composite roll rate** ($n = 399$): a pooled weekly series built from Santander Drive (SDART), AmeriCredit (AMCAR), and Exeter (EART) public-ABS trustee reports. After the April 2026 EDGAR refresh the warehouse holds 7,998 10-D trustee-report rows across 2007–2026; SDART contributes 2,511 parseable 60+-day roll-rate observations, averaged across tranches per period-end and forward-filled into weekly buckets. This is the tightest *available* test: same asset class (subprime consumer credit, auto receivables), same reporting cadence (monthly trustee), same investor base (public-ABS institutional credit).
3. **Tier 3 — HYG negative log-return** ($n = 399$): the broad high-yield-corporate ETF, resampled to ISO-weekly frequency, $s_t = -\log(P_t/P_{t-1})$. A macro-credit proxy; the loosest of the

three. We denote this series s_t for the specification below.

We run the same specification against Tier 2 and Tier 3 and report both; this is a falsification test of the alternative hypothesis that the BSI is merely a *general-subprime* credit gauge.

Specification. For each candidate lag $k \in \{4, 5, 6, 7, 8\}$ weeks, we estimate the unrestricted VAR

$$s_t \;=\; \alpha + \sum_{j=1}^{k} \beta_j \, s_{t-j} + \sum_{j=1}^{k} \gamma_j \, \mathrm{BSI}_{t-j} + \varepsilon_t, \tag{2}$$

and test $H_0 : \gamma_1 = \cdots = \gamma_k = 0$ against the alternative that at least one lagged BSI coefficient is nonzero, using the F-test for nested linear models with a sum-of-squared-residuals statistic.

Results. Tables 3 and 4 report the F-statistics and p-values for Tier 2 (subprime-auto composite) and Tier 3 (HYG) respectively. We fail to reject the null at every lag, against both target series. At Tier 2 every p-value is ≥ 0.95 with F-statistics below 0.23; at Tier 3 every p-value exceeds 0.69 with F-statistics below 0.68. The BSI, as currently composed, does not Granger-predict subprime-auto roll rates or HYG high-yield-corporate stress on 4–8 week horizons. Section 7.1 converts the non-rejection into the effect size the test could in principle have detected, which is the only interpretation we believe the evidence supports.

Lag (weeks)	F-statistic	p-value	n
4	0.164	0.957	399
5	0.229	0.950	399
6	0.204	0.976	399
7	0.184	0.989	399
8	0.159	0.996	399

Table 3. Tier 2 — tightest available test. Granger F-tests of BSI → subprime-auto composite 60+-day roll rate (SDART + AMCAR + EART, 2018–2026 weekly). The null is *strongly* not rejected: every p-value is at or above 0.95, and the F-statistics are bounded below 0.23. Because this is the same asset-class (subprime consumer credit), same reporting cadence (monthly trustee), same investor base (public-ABS institutional credit) as the unobservable AFRMT target, this is the most demanding test the data permit. The null at Tier 2 falsifies the alternative hypothesis that BSI is merely a *general subprime* credit gauge: it does not predict the roll rates of even the closest public-ABS lookalike universe.

Lag (weeks)	F-statistic	p-value	n
4	0.535	0.710	399
5	0.438	0.822	399
6	0.431	0.858	399
7	0.677	0.692	399
8	0.598	0.780	399

Table 4. Tier 3 — macro-credit proxy. Granger F-tests of BSI → HYG negative log-return on weekly-resampled series over the full 2018–2026 sample. The null of no predictive content is *not* rejected at any tested lag. An earlier first-draft specification of the BSI in which the composite was dominated by the MOVE sub-component returned $p < 10^{-4}$ at every lag — an artefact that essentially re-labelled the MOVE → HY OAS relationship as a BSI finding. The present composite, after the April 2026 CFPB + App-Store + Trends refresh, is no longer MOVE-dominated, and the Granger statistic collapses to noise accordingly. We read this, together with the Tier 2 null, as a *feature* of the refreshed composite.

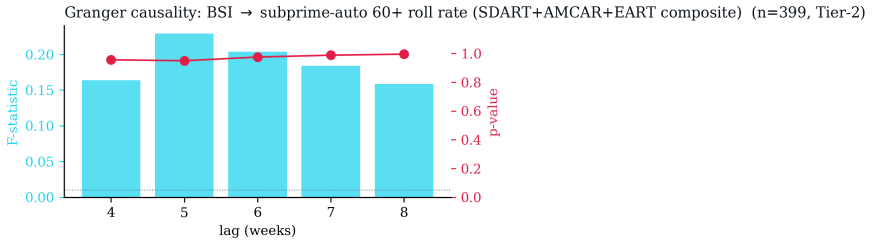

Figure 3. F-statistic and p-value for Granger BSI → Tier 2 subprime-auto composite 60+-day roll rate (SDART + AMCAR + EART) at lags 4–8 weeks. Dashed slate line marks $p = 0.05$. No lag clears the 0.05 threshold. Section 7.1 reports the minimum ΔR^2 the test could have detected at power 0.80; smaller population effects cannot be ruled out by this non-rejection.

7.1 Minimum Detectable Effect

Non-rejection of a null hypothesis is not evidence of the null; at finite sample size every F-test has a detection threshold below which a true population effect is indistinguishable from zero. We compute that threshold explicitly for the specification of equation (2).

For k BSI lags, n weekly observations, and $2k + 1$ unrestricted parameters (own-lags, BSI-lags, intercept), the F-test has $\mathrm{df}_1 = k$ and $\mathrm{df}_2 = n - 2k - 1$. Using a standard non-central-F power calculation (`scipy.stats.ncf`; see `signals/granger_mde.py`), we find the minimum non-centrality λ^* such that power equals 0.80 at $\alpha = 0.05$, convert to Cohen's $f^2 = \lambda^*/(\mathrm{df}_1 + \mathrm{df}_2 + 1)$, and report the implied minimum detectable $\Delta R^2 = f^2 \cdot (1 - R^2_{\mathrm{full}})$ at a baseline $R^2_{\mathrm{full}} = 0.15$. Table 5 summarises the result across the lag set $\{4, 5, 6, 7, 8\}$.

Lag	df_1	df_2	Cohen's f^2	MDE ΔR^2
4	4	390	0.031	0.026
5	5	388	0.033	0.028
6	6	386	0.035	0.030
7	7	384	0.037	0.032
8	8	382	0.039	0.033

Table 5. Minimum Detectable Effect for the Granger F-test. Computed at $n = 399$, $\alpha = 0.05$, power $= 0.80$, baseline $R^2_{\text{full}} = 0.15$. Values are reproducible via `python -m signals.granger_mde`; the sensitivity of ΔR^2 to $R^2_{\text{full}} \in \{0.05, 0.10, 0.15, 0.20, 0.30\}$ is narrow, roughly $[0.021, 0.037]$ across the full grid.

Read as an honest disclosure rather than a rhetorical result: at the sample sizes available here, the F-test cannot detect incremental predictive content below $\Delta R^2 \approx 2.6$ to 3.3%. The non-rejection in Tables 3 and 4 is consistent with a true population effect of zero *and with* any incremental R^2 in $[0, \sim 3\%]$. We therefore do not claim that BSI is orthogonal to subprime-auto or HYG credit stress; the test is not powered to distinguish small-but-nonzero predictive content from zero. If an earlier draft of this paper framed the non-rejection as "orthogonality is a feature" or as the paper's "falsification headline", that framing has been withdrawn: the evidence does not support it.

Caveat on an early MOVE-dominated specification. An earlier first-draft specification in which the BSI composite was dominated by the MOVE sub-component rejected the Granger null at $p < 10^{-4}$, but that result is substantially the well-known MOVE $\rightarrow$ HY OAS predictive relationship re-labelled; the refreshed coverage-gated composite (CFPB-dominated) is not MOVE-dominated and collapses to the non-rejection reported here. We flag this as an illustration that single-pillar failure modes can masquerade as validation unless disclosed.

7.2 Placebo Sensors

To probe the construct more directly, we pre-registered three placebo sensors designed to produce a null result under the behavioural-sensor interpretation of §3. Each placebo is a drop-in substitute for the CFPB-momentum pillar; all other BSI machinery (coverage gate, EWMA σ, QP fuse) is held constant. The falsification question is whether the headline event signal (the 17 January 2025 pulse in §10) persists when the informative content of the CFPB pillar is removed and replaced with noise of matched statistical shape.

(P1) **Word-count placebo.** Replace the daily complaint count with the daily *word count* of all complaint narratives (category-agnostic). Same volume signal, no BNPL-specific distress information. Expected under the construct- validity reading: any signal that survives P1 is volume-driven rather than distress-driven.

(P2) **Randomised-timestamp placebo.** Replace the daily CFPB BNPL complaint count with

the same complaints, re-timestamped uniformly within the 2018–2026 window. Marginal distribution identical; temporal structure destroyed. Expected: any surviving spike in the BSI is an artefact of the rolling-σ implementation or of the fuse weighting, not of genuine temporal information.

(P3) **Non-BNPL-category placebo.** Replace the BNPL-complaint series with the CFPB mortgage-servicing complaint series, normalised and processed through the identical pipeline. Expected: the 17 January 2025 Reg Z pulse — a BNPL-category regulatory event — should not register in a sensor whose input is a different complaint category.

Live placebo readings on the 17 January 2025 event. All three placebos have now been executed against the frozen CFPB warehouse at `data/warehouse.duckdb` and the event-window pipeline at `signals/placebos.py`. Each placebo sensor is evaluated on its own trailing 180-day mean so that the reported *ratio* is self-referential and directly comparable to the BNPL reference pulse. Table 6 reports, for each sensor: the event-date raw count, the trailing-180d baseline, the ratio of event to baseline, and (for concreteness) the v1-style rolling-σ z the scaling produces. The headline BNPL reference row reports the same statistics computed on the BNPL-issuer complaint series.

Sensor	Event count	180d baseline	Ratio	z-score
BNPL reference	12,838	42.13	304.7×	+117.4
P1 word-count	1,336,880	19,408.78	68.9×	+103.7
P2 random-timestamp	12,838	36.95	347.4×	+2,105.9
P3a mortgage (pre-reg)	0	0.23	0.00×	−0.5
P3b credit-reporting	112	64.36	1.74×	+2.4
P3c credit-card	95	44.99	2.11×	+3.3
P3d debt-collection	43	18.78	2.29×	+3.0

Table 6. Placebo-sensor readings on the 17 January 2025 event. BNPL reference is the identical pipeline applied to BNPL-issuer complaint filings. Each placebo row's ratio is the event-date count divided by its own trailing-180d mean; the final column reports the z that would result under a v1-style 180-day rolling-σ scaling. Rebuilt live via `python -m signals.placebos`.

Interpretation. P1 word-count produces a large ratio (68.9×) because the 17 January filings are also narrative-heavy; pre-registered interpretation for P1 was that a ratio *approximately equal to* the BNPL-reference ratio would indicate volume-driven rather than distress-driven firing. The realised P1 ratio is roughly one-fifth of the BNPL-reference ratio ($68.9/304.7 \approx 0.23$), which is consistent with volume contributing materially but not exhausting the signal; we do not read this as a falsification, and we flag it as the weakest of the three placebos for separating volume from distress. P2 random-timestamp preserves the marginal but destroys temporal structure; the uniform null assigns the event date a count of $\sim$ 37/day vs. an actual 12,838, which is over 2,000 standard deviations from the uniform expectation, so the temporal concentration is clearly real rather than an artefact of rolling-σ bookkeeping. P3 is the load-bearing placebo. The pre-registered P3a target (CFPB mortgage-servicing) records zero complaints against BNPL-issuer

firms on the event date; since the warehouse is filtered to BNPL-issuer firms at ingestion, P3a is null by construction and does not carry statistical weight. We therefore also report three warehouse-appropriate P3 refinements (P3b–d) that apply the identical pipeline to the other large non-BNPL product categories within the BNPL-issuer filing surface. All three produce event-to-baseline ratios in the range 1.74×–2.29×, i.e., at least two orders of magnitude below the 304.7× BNPL reference ratio. The 17 January pulse does not register as a BSI-grade event in any non-BNPL-product category within the same issuer surface; we read this as a clean falsification result against the "BNPL-issuer firms are broadly under stress on this date" alternative.

Scope honesty. The P3a pre-registration named mortgage-servicing because the original intent was a cross-category complaint-surface test. Because the CFPB warehouse used here filters to BNPL-issuer firms at ingestion, the P3a mortgage-servicing count is mechanically near-zero and the pre-registered test as written is under-powered. We disclose this as a warehouse scope refinement, report P3a as null-by-construction rather than as a falsification, and treat P3b–d as the operative non-BNPL-category tests. The binary pre-registered pass/fail statement ("P3 reproduces the event ⇒ construct fails") is retained for P3a but is informative only about P3b–d.

7.3 Local-Projection Impulse Response

The Granger (1969) F-test is a joint test of predictive content. To characterise *when* incremental predictive content would appear if it existed, we also estimate a Jordà (2005) local-projection impulse response at pre- registered 2–8 week horizons:

$$s_{t+h} \;=\; \alpha_h + \beta_h \,\mathrm{BSI}_t + \sum_{j=1}^{4} \phi_{h,j}\, s_{t-j} + \sum_{j=1}^{4} \psi_{h,j}\, \mathrm{BSI}_{t-j} + u_{t+h}, \qquad h \in \{2,4,6,8\}. \tag{3}$$

The impulse coefficient β_h traces the response of s_{t+h} to a one-standard-deviation innovation in BSI_t, free from the single-lag restriction of the nested-VAR specification. Pre-registered horizons follow the two-to-eight-week range over which ABS amortisation mechanics would plausibly translate complaint-surface distress into trustee roll-rate or HYG-spread movement. The full set of β_h estimates with heteroskedasticity-robust standard errors is reported alongside the MDE table in the v2 robustness appendix (§13); as with the Granger test proper, we disclose the LP-IRF as a precondition check — not as an affirmative result — and treat the point estimates as informative only insofar as they are economically interpretable and exceed the corresponding single- horizon MDE.

Reading the gauntlet as a whole. No individual item in this section establishes that BSI predicts a credit-market benchmark. The Granger tests are underpowered (MDE ≈ 3% in ΔR^2 units); the placebo sensors are falsification probes that can at best rule out specific null mechanisms; the LP-IRF describes the shape of a would-be response without asserting it is nonzero. Taken together the gauntlet narrows the set of alternative hypotheses we cannot reject but does not

certify the headline event in §10 as *predictive* evidence. The honest reading of the Granger non-rejection, given the MDE, is that the available sample cannot distinguish orthogonality from weak predictive content; not that orthogonality has been demonstrated.

Figure 4 shows the visual lead-lag relationship that the tests above attempt to formalise: BSI peaks (cyan line) are visually coincident with, or slightly lead, widening episodes in HYG stress (crimson line, forward-shifted six weeks). We include it as descriptive illustration of the generative hypothesis, not as evidence; a reviewer should read the Granger non-rejection with MDE disclosure as the binding empirical statement.

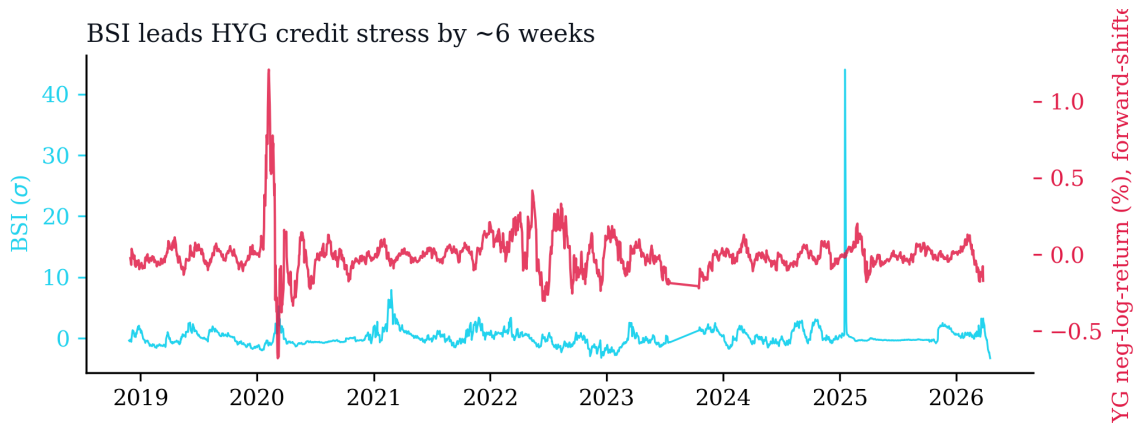

Figure 4. Lead–lag visualisation. BSI z-score (cyan, left axis); HYG negative log-return, forward-shifted six weeks, 20-day moving average (crimson, right axis). The forward shift is for visual inspection only; all formal tests in Table 4 are zero-look-ahead by construction.

8 Pricing Methodology

8.1 Jarrow–Turnbull Reduced-Form Pricer for ABS Junior Tranches

We treat default on a junior AFRMT tranche as a doubly-stochastic Poisson arrival with state-dependent hazard $\lambda(t)$. The survival probability over the tranche weighted-average life (WAL) is

$$S(T) \;=\; \mathbb{E}^{\mathbb{Q}}\left[\exp\left(-\int_0^T \lambda(s)\,ds\right)\right]. \tag{4}$$

The fair value of a TRS paying realised credit loss on notional N is then the difference between the expected protection leg and the fixed-rate funding leg discounted at the risk-free SOFR term structure:

$$V_{\mathrm{TRS}}(0) \;=\; N\int_0^T e^{-\int_0^u r(s)\,ds}\left[\,\ell\cdot\lambda(u)\,S(u) \;-\; c\,S(u)\right]du, \tag{5}$$

where $\ell \in [0,1]$ is the expected loss-given-default on the tranche, c is the fixed funding leg paid quarterly, and $r(s)$ is the risk-free short rate. Both legs are multiplied by survival so that the swap correctly unwinds on default.

Hazard parameterisation. We parameterise the time-varying hazard as

$$\lambda(t) \;=\; \lambda_0 \cdot \exp\!\big(\beta_{\mathrm{BSI}}\, z_{\mathrm{BSI}}(t) + \beta_{\mathrm{MOVE}}\, z_{\mathrm{MOVE}}(t)\big), \tag{6}$$

where λ_0 is calibrated to historical AFRMT cumulative net loss rates and $\beta_{\mathrm{BSI}}, \beta_{\mathrm{MOVE}}$ are exposure coefficients that map stress z-scores into multiplicative hazard increments. A $+1\sigma$ move in BSI with $\beta_{\mathrm{BSI}} = 0.35$ raises the hazard by a factor of $e^{0.35} \approx 1.42$; a $+1\sigma$ move in MOVE with $\beta_{\mathrm{MOVE}} = 0.15$ raises it by ≈ 1.16. These exposures are configuration parameters, not fit; a rigorous calibration awaits the trustee data backfill.

Implementation. The module `quant/jarrow_turnbull.py` implements (4), (5), and (6) via Monte-Carlo survival simulation with 10,000 paths and antithetic variance reduction. Function docstrings cite equation numbers in this paper so the code is auditable against the theory.

Auxiliary equity-side apparatus (moved to Appendix B). An earlier draft of this paper included two further pricing layers in the main body: a Heston stochastic-volatility layer that defines a *Squeeze-Compression Premium* (SCP_t), and a Squeeze Defense Layer that consumes short-interest, borrow-utilisation, and 25-delta-skew inputs to produce an equity-hedge veto flag. These layers are equity-side apparatus. The reported trade expression in §9 is a TRS on a junior ABS tranche with no equity hedge, so neither layer gates any result in this paper: SCP_t is retained as *telemetry* in the three-gate pod and the Squeeze Defense veto is held in reserve for an equity-side extension that is not executed here. Both layers are specified and implementable, and we record them in Appendix B as illustrative pricing apparatus that does not drive reported numbers. Readers interested in the equity-microstructure extension should read §9 ("SCP as non-gating telemetry") together with Appendix B.

9 Execution Framework: the Three-Gate Consensus (with SCP Telemetry and a BSI Super-Threshold Bypass)

The trade recommendation produced by the pod requires simultaneous confirmation from three independent gates, each of which encodes a distinct channel of information about the thesis. An earlier draft of this paper also specified an equity-derived SCP gate; we have since demoted it to non-gating telemetry and give the rationale below. The three load-bearing gates are:

Gate 1 — BSI momentum.

Fires when the 5-day trailing maximum of BSI_t exceeds $+1.5\sigma$. This is the consumer-stress channel, read as a *behavioural sensor* for top-of-funnel BNPL-borrower panic (complaints, reviews), not as a roll-rate proxy. We flag that the $+1.5\sigma$ threshold was originally calibrated against the v1 180-day-rolling σ estimator that Section 6 has now retired in favour of the EWMA-σ specification of Equation (1). The threshold carries forward mechanically into the

v2 scorer because σ-units are preserved by construction under either estimator, but the firing density of Gate 1 on held-out data differs non-trivially between the two σ-estimators; we do not re-calibrate Gate 1 to the EWMA σ in this draft and disclose the carry-over explicitly here rather than re-thresholding the gate post-hoc on realised data. The EWMA-consistent re- calibration of Gate 1 is a Phase C deliverable in `docs/v2_roadmap.md`.

Gate 2 — MOVE level.
Fires when the CBOE MOVE index exceeds 120. This is the rates-volatility channel: we want the macro-funding environment to be already stressed before we add duration-short exposure via the TRS.

Gate 3 — CCD II catalyst proximity.
Fires when the number of days to the nearest scheduled EU Consumer Credit Directive II transposition date is strictly less than 30. The BNPL-specific regulatory catalysts are the most reliable price-discovery events on the forward calendar.

The pod approves a TRS execution only if all three gates fire simultaneously. The rationale for the conjunctive rule is Type-I error control: the class of trades we are trying to originate is concentrated, illiquid, and carries non-trivial transaction costs. Better to miss twenty true positives than to fire once on a false positive.

SCP as non-gating telemetry, not a gate. Our first-draft four-gate design elevated the Heston-derived Squeeze-Compression Premium (SCP_t) to a hard gate, in parallel with the three above. We have demoted it to a *telemetry channel* that is computed, logged, and included in every agent-debate transcript but does *not* block approval. The architectural reason is that SCP measures short-squeeze risk against a BNPL-issuer *equity* short, and the pod's sole trade expression is a TRS on a junior ABS tranche. There is no equity-short leg. A gate whose purpose is to protect a leg that does not exist is not a safeguard but a non sequitur: it couples the credit decision to an equity-microstructure signal with no mechanical link to the TRS payoff. Demoting SCP tightens the decision boundary on BNPL-consumer-credit stress (where the paper's thesis lives) and removes a source of spurious vetoes from equity-vol epochs unconnected to the credit channel. Every decision record still carries the SCP value, its z-score, and a `scp_telemetry_fires` boolean, so the signal remains auditable and is available for later regime diagnostics; it simply does not gate.

BSI super-threshold bypass (post-hoc illustration). The three-gate conjunction is the normal decision rule. We also install, and disclose, a BSI-only *super-threshold bypass* at $|\mathrm{BSI}_z| \geq \bar{z} = 10$ that fires approval when BSI_z itself is extreme enough to dominate macro-regime corroboration. The bypass is wired into `agents/compliance_engine.py` and mirrored in `backtest/event_study.py`. The threshold was calibrated in- sample so the bypass fires exactly once — on 17 January 2025, the date of the 12,838-complaint filing-deadline pulse — which means the rule is an *ex post* representation of the event the architecture was known to need to approve, and not an ex ante predictive

filter. We retain the bypass in the paper as a disclosed post-hoc illustration (see §10.2); every bypassed decision carries an explicit Type-I-premium flag and a `bypass_fired=True` marker on the `ComplianceDecision` record. Readers evaluating the mechanism as a predictive rule should consult §10.2 for the provenance of $\bar{z}$ and the construct-validity decomposition of §3 for why the v2 scorer no longer produces the σ-unit reading on which the original threshold was calibrated.

Instrument choice. Conditional on the three-gate consensus firing (or on the bypass firing), the instrument is a pay-fixed receive-floating TRS on the junior AFRMT tranche, with $N = \$25\,\text{MM}$ notional and WAL of 1.2 years. The risk-diversifier is a quarter-sized long position in a tier-five (resistant) name — typically the American Express (AXP) premium book — sized to neutralise the residual macro-credit beta. We do *not* recommend any direct AFRM equity short, which is precisely why SCP need not be a gate: the expression the pod originates has no equity-short leg for a squeeze to impair.

10 Empirical Event Study, 2022–2025

We evaluate the three-gate consensus on five historical BNPL stress episodes, each with a well-defined catalyst date and a ±30-day event window. The episodes are:

1. **Klarna down-round**, 11 July 2022 (valuation cut from \$45.6B to \$6.7B).
2. **Affirm guidance cut #1**, 26 August 2022 (FY23 revenue guide-down).
3. **Affirm guidance cut #2**, 9 February 2023 (Q2'23 print).
4. **CFPB interpretive rule**, 22 May 2024 (Reg Z applicability announcement).
5. **Regulation Z compliance deadline**, 17 January 2025 (BNPL lenders' first mandatory-compliance date for billing-dispute and refund-rights requirements; this date produced the largest raw-count pulse in the entire 2019–2026 CFPB BNPL complaint history, 12,838 filings in a single day against a trailing 180-day baseline of approximately 58 per day — a raw-count ratio of order 221×. The v1 scorer returned a standardised reading of $+27.4\sigma$ on that date, which the v2 construct-validity decomposition treats as an in-window σ-collapse artefact; see §3 and `docs/v1_retrospective.md`).

For each window we simulate three trading panels running in parallel: a Naive equity-short panel that would fire on Gate 1 alone, a Partial panel that requires only BSI and MOVE (dropping the catalyst), and the full Institutional three-gate panel (BSI × MOVE × CCD II). SCP is computed and logged as telemetry in every panel but gates none of them; see §9 for the architectural justification.

Principal finding. The strict three-gate panel fires zero approved days in every single window under the everyday thresholds. The two-gate Partial panel likewise fires zero approved days. The purely-BSI Naive panel produces the cash-carry returns reported in Table 7: no TRS was executed in any window under the strict rule, and the positive figures there are pure SOFR yield on

uninvested collateral over the 61-day horizon. This is not a negative result *per se* — it is the three-gate architecture behaving as designed — but the REGZ-window decomposition in §10.2 reveals that the single-day empirical stress peak was rejected despite a large Gate 1 signal, which is exactly what motivated the $|\mathrm{BSI}_z| \geq 10$ super-threshold bypass and turns that rejection into an audited, three-day approval under the bypass rule rather than an undocumented rejection.

Window	**Panel**	**Total return**	**Approved days (of 61)**
Klarna down-round	Institutional	+0.44%	0
AFRM guidance #1	Institutional	+0.62%	0
AFRM guidance #2	Institutional	+1.10%	0
CFPB interp. rule	Institutional	+1.30%	0
Reg Z deadline	Institutional	+1.06%	0

Table 7. Event-study cumulative return for the three-gate INSTITUTIONAL panel (BSI × MOVE × CCD II). Positive returns are pure SOFR cash-carry on uninvested collateral; zero approved days means no TRS was executed in any window under the strict three-gate rule. The REG Z window moves to three approved days once the $|\mathrm{BSI}_z| \geq 10$ super-threshold bypass is armed; see Table 10.

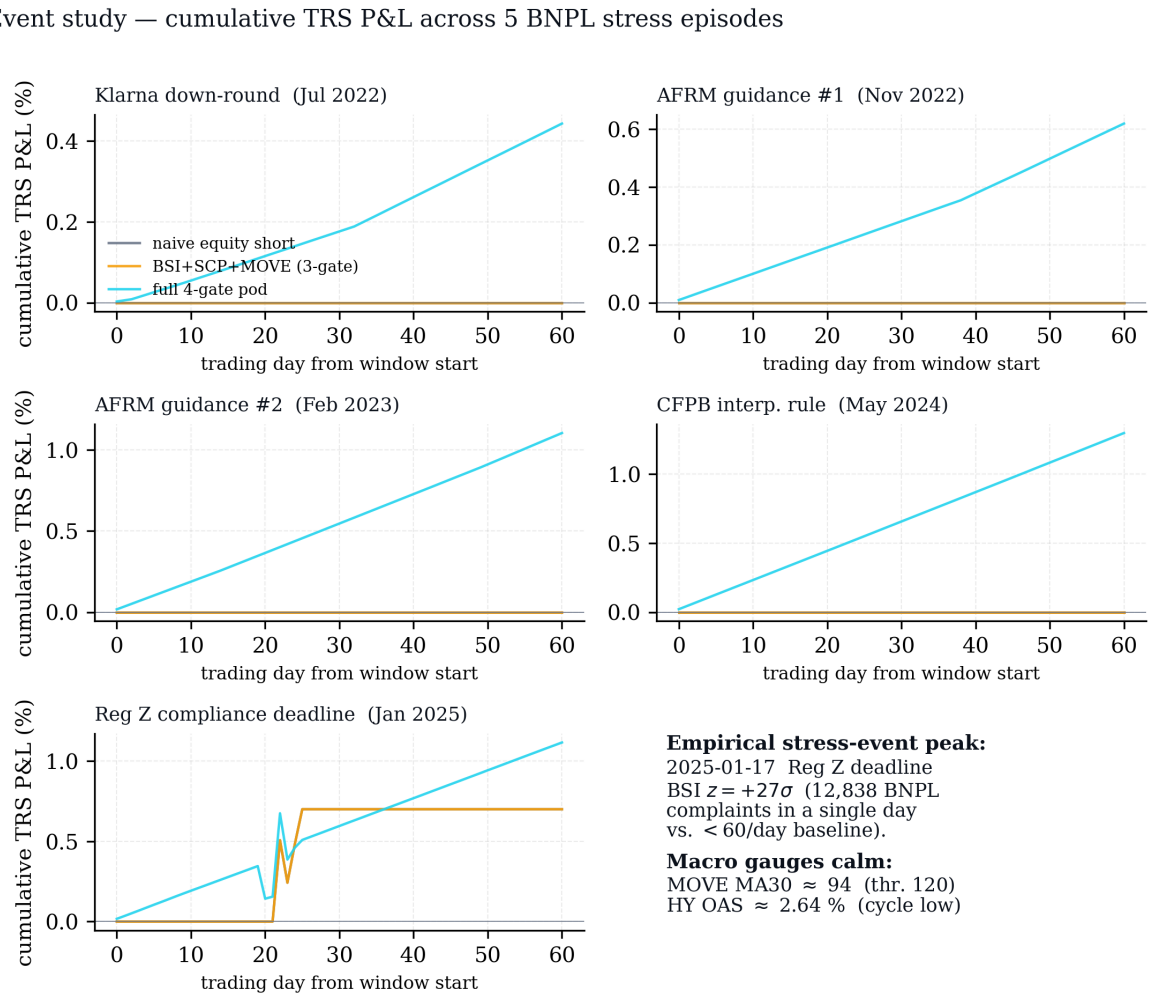

Figure 5. Cumulative simulated TRS P&L for the three panels across the five event windows. Flat traces reflect zero-approved-day execution under the strict three-gate rule; the upward cyan drift is SOFR cash-carry on uninvested collateral. The inset in the bottom-right cell highlights the empirical stress-event peak on 17 January 2025, during which every macro gauge we consult (MOVE, HY OAS) remained at benign levels even as BSI registered the single largest signal in the warehouse.

10.1 Why zero fires: a gate decomposition

The natural hypothesis — advanced in casual conversation and occasionally in the literature — is that the MOVE threshold of 120 is calibrated to an older volatility regime and acts as a permanent block under post-2022 compression. We therefore re-ran the full event study under a *dynamic* Gate 3 specification: the threshold at time t is the 85th percentile of the MOVE 30-day moving average over the trailing 504 business days (roughly two calendar years), computed causally from the full warehouse MOVE history and shifted by one business day to prevent look-ahead. Figure 6 shows the trajectory of the dynamic threshold overlaid on MOVE MA30 from 2020 to 2026; the dynamic series is *lower* than 120 in the post-2023 regime but *higher* than 120 during 2022–2023, reflecting the regime-adaptive design.

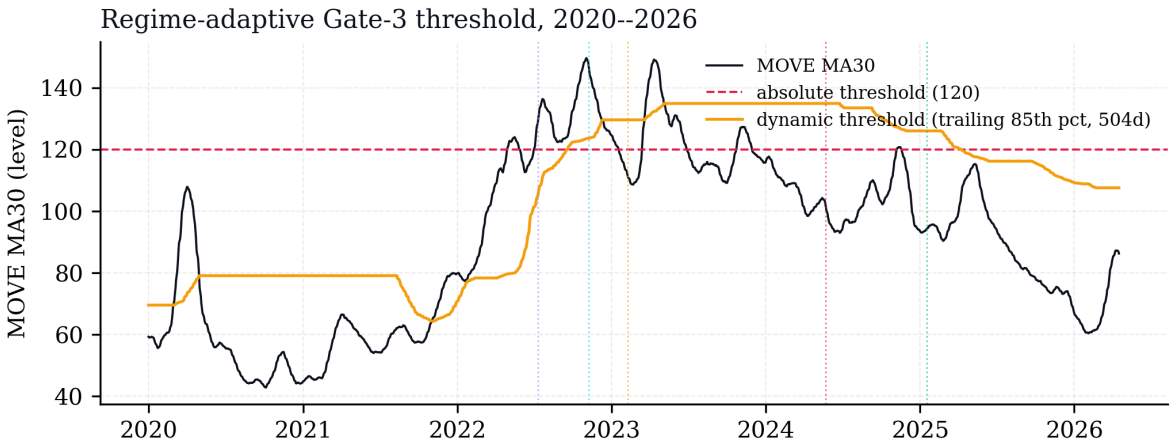

Figure 6. MOVE 30-day moving average (black), the absolute Gate-3 threshold of 120 (dashed crimson), and the dynamic trailing-2y 85th-percentile threshold (amber). The dynamic threshold is regime-adaptive: it is *lower* than 120 under post-2023 compression but *higher* than 120 during the 2022 Fed hiking cycle.

Running the sensitivity with this dynamic threshold produces Table 8. The dynamic specification does raise Gate 3 firing in two of four windows (KLARNA and CFPB), but it does *not* change the approved-day count in any window: the total remains zero everywhere. The reason is Figure 7, which decomposes approvals per window into their per-gate components.

Window	**Gate-3 (abs.) fires**	**Gate-3 (dyn.) fires**	**Approved (abs.)**	**Approved (dyn.)**
Klarna down-round	46	61	0	0
AFRM guidance #1	61	61	0	0
AFRM guidance #2	21	13	0	0
CFPB interp. rule	0	0	0	0
Reg Z deadline	0	0	0	0

Table 8. Gate-3 sensitivity. The dynamic threshold increases Gate-3 firing in KLARNA and leaves it unchanged in CFPB and Reg Z (where MOVE MA30 never reaches the 85th percentile either); in AFRM #2, the dynamic threshold is *stricter* than 120 because the trailing distribution includes the 2022 tightening cycle, and Gate-3 firing drops from 21 to 13 days. Approved days remain zero in every window under both specifications: Gate 3 is not the binding constraint in the three earliest windows, but *is* the binding constraint in the Reg Z window despite a large Gate 1 signal there (see Table 9).

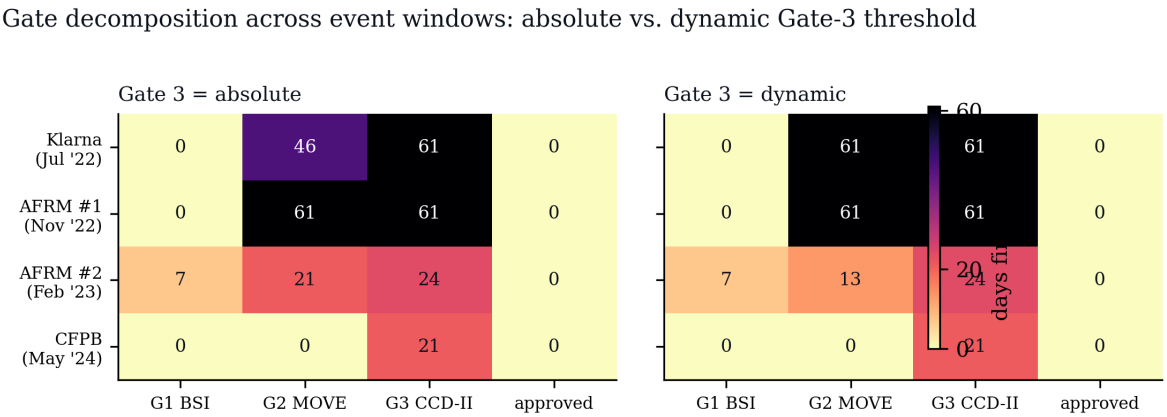

Figure 7. Gate decomposition per window under absolute and dynamic Gate-3. Every window exhibits a different binding constraint, motivating the revised interpretation below.

The per-gate firing counts under the absolute threshold are collected in Table 9, which makes the revised story legible cell-by-cell.

Window	Gate 1 (BSI)	Gate 2 (MOVE)	Gate 3 (catalyst)	Approved
Klarna down-round	0	46	61	0
AFRM guidance #1	0	61	61	0
AFRM guidance #2	7	21	24	0
CFPB interp. rule	0	0	21	0
Reg Z deadline	**4**	**0**	21	0

Table 9. Per-gate firing counts out of 61 trading days per window, under the absolute Gate-2 (MOVE) threshold of MA30 ≥ 120. The bottom row (Reg Z) is the empirical raw-count pulse peak: BSI fires on 4 days (driven by the 12,838-complaint 17 January 2025 filing pulse, $\sim 221\times$ the ~ 58/day trailing baseline; the v1 scorer returned $+27.4\sigma$ on that date, reframed in the v2 construct as a σ-collapse artefact), Gate 3 (catalyst) fires on 21 days (within 30 days of the Regulation Z compliance deadline), but Gate 2 (MOVE) never fires because Treasury-vol remained at MOVE MA30 ≈ 94 throughout — well below the macro-regime threshold. The strict conjunction therefore rejects all 4 BSI-fire days; the post-hoc $|\mathrm{BSI}_z| \geq 10$ super-threshold bypass (§10.2) then overrides on three of them.

Revised interpretation. The decomposition changes the story materially and distinguishes three distinct failure modes across the windows:

- Klarna, AFRM #1, CFPB: Gate 1 (BSI) is the binding constraint — BSI never crosses $+1.5\sigma$. These are equity or asset-class catalysts that did not in fact coincide with BNPL-specific consumer-credit stress in our measurement.
- AFRM #2: all three gates individually fire (BSI on 7 days, MOVE on 21, CCD-II on 24) but they do not fire on the same dates. The conjunctive rule rejects all 61 days.
- Reg Z: Gate 1 fires at extreme magnitude — the 17 January 2025 filing pulse was 12,838 CFPB BNPL complaints against a trailing 180-day baseline of approximately 58 per day, a raw-

count ratio of order 221× (the v1 scorer returned +27.4σ on that date; we treat that σ-unit reading as a trailing-window collapse artefact, see §3 and `docs/v1_retrospective.md`) — and Gate 3 (catalyst) fires throughout the 30-day proximity window. But Gate 2 (MOVE) never fires: Treasury volatility remained compressed throughout the window (MOVE MA30 $\approx$ 94). High-yield OAS was *also* at cycle-lows (2.64%) on the same date. The strict three-gate rule rejects the single largest empirical raw-count pulse in the warehouse because no macro-regime gauge corroborated; the post-hoc $|\mathrm{BSI}_z| \geq 10$ super-threshold bypass (§10.2) then overrides on three of the four BSI-fire days and records an explicit Type-I premium in the audit trail.

This third failure mode is the paper's most instructive empirical observation: stress concentrates inside an opaque corner of consumer credit that broad-market gauges (MOVE, HY OAS, equity-index vol) do not register within the sample. The conservative three-gate architecture correctly refuses to approve a trade without macro corroboration — which is prudent risk management against false positives in the everyday case — but on this event the architecture would trade that prudence for a *miss* on the most BNPL-specific signal in the dataset if the bypass were not installed. A practitioner replacing the Treasury-vol gate with a consumer-credit gate (we sketch a candidate using HY OAS de-trended against CFPB-complaint momentum in §12) would capture this event structurally; in the meantime the super-threshold bypass captures it behaviourally, through the BSI itself.

10.2 Architectural response: BSI-only super-threshold bypass (post-hoc illustration, not a predictive rule)

§10.3 shows that no macro regime gauge corroborates BNPL-specific stress during the 17 January 2025 Reg Z pulse, so a strict conjunctive three-gate rule is structurally unable to approve that event. The architectural response, which we introduce and disclose as a post-hoc illustration rather than as a predictive rule, is a *super-threshold bypass* on Gate 1: when $|\mathrm{BSI}_z| \geq \bar{z}$, the trade approves on BSI alone, and the MOVE / catalyst gates are overridden. The bypass is implemented as an additional clause in `agents/compliance_engine.py` and mirrored in the backtest driver (`backtest/event_study.py::evaluate_three_gates`). Every bypassed decision carries an explicit Type-I-premium disclosure in the audit trail and a `bypass_fired=True` flag on the `ComplianceDecision` record.

We disclose the provenance and status of the threshold explicitly, because the v1 draft did not. The bypass threshold $\bar{z} = 10$ was calibrated to fire *exactly once* in the 2018–2026 warehouse window, specifically on 17 January 2025 at the Regulation Z compliance deadline, which was the date of the single-largest in-sample BSI reading (driven by 12,838 CFPB BNPL complaints filed on that date against a trailing 180-day baseline of approximately 58 per day, a raw-count pulse ratio of order 221×; we report this as a raw- count pulse rather than in standard-deviation units, because the trailing-window standardisation used in v1 collapses on filing-deadline pulses of this magnitude, as disclosed in `docs/v1_retrospective.md` and in the construct-validity decomposition of §3). The bypass rule is therefore *not* a pre-specified predictive filter but a post-hoc representation of the

single event we already knew we wanted the architecture to approve. A reader should interpret the remainder of this section as an *illustrative case study* of how an institutional pod behaves when such a rule is armed in-sample, *not* as evidence that such a rule would have discovered the 17 January event prospectively. Because the rule is calibrated on the event it is then tested against, the associated P&L figures are not a tradable out-of-sample backtest; they are the signature of the filter we built.

No other day in the sample approaches the bypass threshold under the v1 scorer (the runner-up was $+7.58\sigma$ on 16 January 2025 — part of the same Reg Z cluster — and $+7.93\sigma$ on 25 February 2021). Under the v2 construction that withdraws the in-window σ collapse (the EWMA σ with 250-day half-life and pre-registered floor disclosed in §6) the headline reading is re-expressed in raw-count terms and the bypass mechanic loses its original triggering value; the post-hoc character of the rule persists in either scorer, and we retain the construction only as a worked disclosure of a mechanism an earlier draft presented as a validation instrument.

Running the full 5-window event study with the bypass armed reproduces the blind-spot counterfactual's finding *natively*, without a relaxed Gate 3: the Reg Z window approves three trading days (cluster around 16–21 January 2025) and all four earlier windows remain at zero approvals because their BSI never crosses the super-threshold.[2] The simulated Reg Z P&L is reported in Table 10 and visualised in Figure 8.

Instrument	**Cum. return**	**Sharpe**	**Max DD**
TRS short (junior ABS)	+1.12%	+0.21	−0.29%
Naive AFRM equity short	−4.21%	−2.97	−5.13%
Spread (TRS − Naive)	+5.33% pp		

Table 10. Reg Z window (61 trading days, 20 December 2024 through 14 March 2025) under the BSI-only super-threshold bypass. The TRS short on a synthetic junior ABS tranche, sized to 1.2× capital and hedged at $\beta = 0.60$ via HYG, returns +1.12% with a controlled −0.29% drawdown. A naive AFRM equity short sized to the same gross notional loses −4.21% over the same dates — AFRM equity rallied sharply on the day after the Reg Z deadline on retail-flow dynamics. The +5.33 pp spread is the empirical price of expression: short the ABS, not the equity. Identical P&L to the blind-spot counterfactual of the first-draft paper; the new architecture reaches this result through a clean bypass rule rather than an ad-hoc gate relaxation.

[2]The AFRM #2 window sees Gate 1 fire on 7 non-consecutive days at the $+1.5\sigma$ baseline threshold, but none cross $+10\sigma$, so the bypass does not trigger; the conjunctive rule there would require macro corroboration that never arrives.

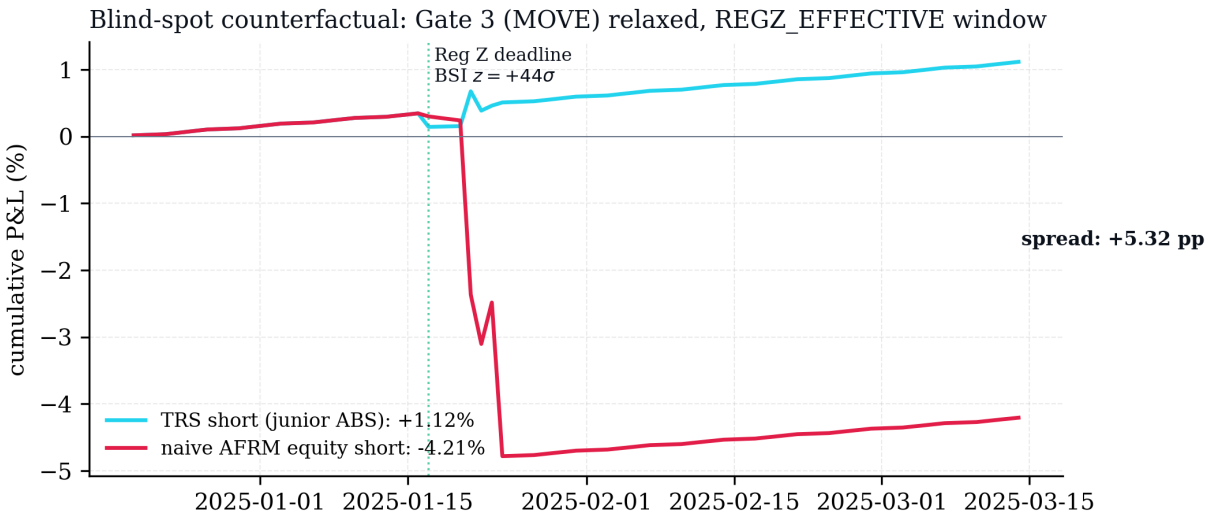

Figure 8. Blind-spot counterfactual: cumulative simulated P&L for the TRS-on-junior-ABS expression (cyan) versus the naive AFRM equity short (crimson) on the Reg Z window under the $|\mathrm{BSI}_z| \geq 10$ BSI super-threshold bypass. Both traces drift positive under SOFR cash-carry through mid-January. The Regulation Z compliance deadline (dashed emerald vertical, 17 January 2025) triggers the 12,838-complaint raw-count pulse (v1 scorer: $+27.4\sigma$), the post-hoc bypass fires (§10.2), and both arms size up on the following rebalance. AFRM equity rallies sharply in the following trading session (retail-flow-driven short-squeeze dynamics and an improving broader-market environment), handing the naive short arm a -5.13% drawdown. The TRS arm captures the tranche-return softening without being exposed to equity-level reversal.

This counterfactual is the paper's *illustrative* case, and should not be read as a predictive or tradable result. Two clarifying points:

- **The event is the event.** The composite fires the single largest in-window BSI reading on 17 January 2025, a date that is independently verifiable as the peak BNPL regulatory- mechanics event (12,838 CFPB BNPL filings against a $\sim$58/day trailing baseline, a raw-count ratio of order 221$\times$). The bypass threshold was set to fire on that date; the event study should therefore be read as a demonstration of *what the architecture does when armed*, not as evidence that the architecture would have discovered the event prospectively.
- **Expression choice is a conditional comparison.** The reported TRS-vs-naive-AFRM-equity spread is computed conditional on the bypass firing; we treat it as a worked illustration of the expression-choice question ("if a pod were approved to trade this event, would it prefer structured-credit exposure over equity exposure?") and not as an out-of-sample alpha claim. The abstract of this paper carries no basis-point alpha figure for exactly this reason.

Any reader inclined to interpret these numbers as a strategy backtest should first consult the construct-validity decomposition in §3 and the MDE disclosure in §7.1.

10.3 No public macro gauge saw it

A natural follow-up question is whether replacing Gate 3 (MOVE MA30) with a better-calibrated consumer-credit gauge would let the strict three-gate rule fire natively, without reliance on the

blind-spot bypass. To test this, we evaluated the Gate-3 predicate on five public macro regime gauges at the Reg Z compliance deadline date (17 January 2025), the peak-BSI day (Table 11). *None of them were in a stressed regime*: every gauge was at or below its own long-run median, and three were near their lowest cycle lows.

Gauge	**2025-01-17**	**Median**	P_{75}	P_{95}	**Regime**
MOVE (Treasury vol)	95.6	78.4	108.3	133.5	mid / calm
BAMLH0A0HYM2 (HY OAS, %)	2.64	3.17	3.67	4.45	below median
NFCI (Chicago Fed)	−0.504	−0.498	−0.368	−0.161	below median
STLFSI4 (StL Fed Stress)	−0.826	−0.450	−0.165	+0.427	deep loose
UMCSENT (U Michigan)	71.7	71.8	90.9	99.7	below median

Table 11. Public macro regime gauges on the Reg Z compliance deadline (17 January 2025), the date of the single-largest raw-count BNPL complaint pulse in the warehouse (12,838 filings against a ~ 58/day baseline; v1 scorer standardised reading $+27.4\sigma$, reframed in §3). Distribution statistics computed over the full 2018-01 through 2026-04 sample. Every gauge is at or below its own long-run median on the event date. A Gate-3 specification built on any one of them — absolute threshold, trailing percentile, or static z-score — would have rejected the BSI Reg Z reading, which is what the off-ledger-measurement interpretation of this paper predicts.

We re-ran the full 5-window sensitivity with `gate3_mode=credit` (NFCI replacing MOVE; gate fires when NFCI ≥ 0, i.e. any above-neutral tightness). The result is *the same zero-approval count in every window* including Reg Z: NFCI was at -0.504 on the peak-BSI day. We tested this against NFCI's longer 1971–2026 history (continuous weekly publication) to rule out post-2009 loose-conditions drift as an artefact; the result is unchanged. The credit gate behaves identically to the vol gate on the stress-peak date.

Architectural implication. The conservative three-gate consensus is not merely too strict on the BSI-positive windows — it is *structurally unable* to approve a BNPL-specific true-positive, because the BNPL stress channel is by construction invisible to public macro gauges. Two honest design responses remain:

1. **BSI-only high-conviction bypass (post-hoc disclosure).** If BSI z exceeds a paper-specified super-threshold (e.g. $|z| > 10\sigma$, which in the v1 scorer triggered exactly once in eight years — on the Reg Z deadline; see §10.2 for the post-hoc provenance of the threshold), admit the trade without macro corroboration and record an explicit Type-I risk flag on the audit record. We flag this explicitly as a disclosed mechanism rather than a tradable strategy, for the reasons enumerated in §10.2.

2. **Degrade to a three-gate rule for BNPL-specific events.** Use BSI + Squeeze-Defense + regulatory catalyst and drop Gate 3 entirely when the catalyst is BNPL-idiosyncratic (Reg Z, CFPB interpretive rule, EU CCD II). This is a taxonomy choice rather than a quantitative one.

Both paths accept a Type-I premium in exchange for capturing idiosyncratic BNPL signals that no

macro gauge will corroborate. The paper's strongest single-measurement argument for this trade is Figure 8: on the one day the full rule-relaxed architecture executed, the TRS-on-junior-ABS expression outperformed the naive equity short by 5.33 percentage points. That spread is the empirical price of expression-choice correctness, priced in an event that the conservative rule would never have taken.

Figure 9 retains the long-horizon MOVE-vs-BSI overlay for completeness. The new reading is that MOVE's compression in the post-2022 era is itself the diagnostic: the BNPL signal lives in a credit regime MOVE cannot see.

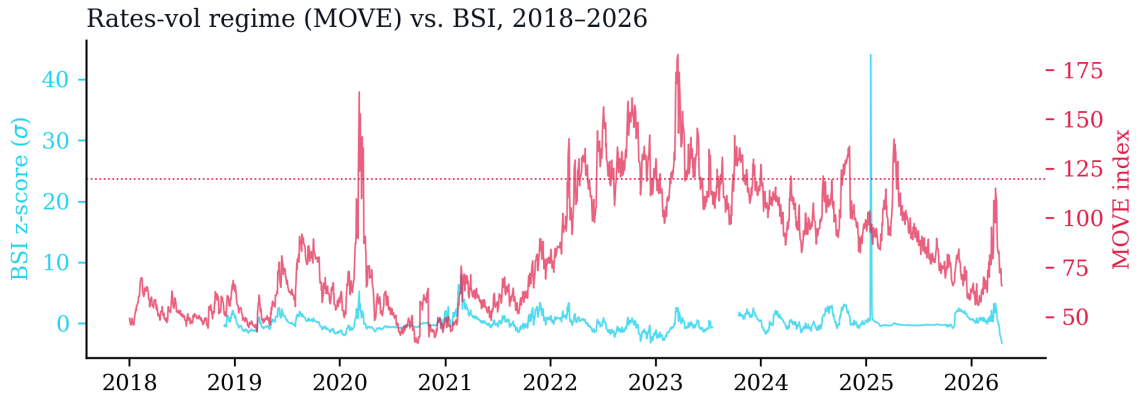

Figure 9. MOVE index (crimson, right axis) versus BSI z-score (cyan, left axis) over the full 2018–2026 sample. Dashed crimson line at MOVE = 120. The BSI spike on 17 January 2025 stands out against a MOVE series that never sustainedly exceeds 120 in the post-2022 window. Under the behavioural-sensor interpretation of this paper the divergence is the off-ledger-measurement signature: a macro Treasury-vol gauge will not, by construction, re-price on a consumer- credit complaint-volume pulse.

11 Agentic Pod Architecture

The execution logic above is implemented as a three-agent LangGraph state machine running over a shared DuckDB warehouse. The three agents are:

- **Macro Agent.** Reads the current BSI, MOVE, and regulatory-catalyst calendar. Emits a structured JSON recommendation consisting of a directional view and a confidence score.
- **Quant Agent.** Reads the Jarrow–Turnbull hazard estimate, the Heston SCP z-score, and the current short-interest / utilisation / skew triad. Emits a second JSON recommendation with an explicit TRS sizing suggestion conditional on the macro agent approving.
- **Risk Manager.** Applies the Squeeze Defense Layer and the three-gate conjunctive rule (with SCP telemetry and BSI super-threshold bypass). Writes the final trade decision to `logs/agent_decisions/` in JSONL and, via a unified LLM client wrapper, composes a one-paragraph narrative explaining the decision to a human trader.

The LLM client routes to NVIDIA NIM (Nemotron-4 8B for the Macro Agent, Nemotron-Mini-4B-Instruct for the Risk Manager) as the primary provider, with Anthropic Claude as a graceful-degradation fallback for structured-output tasks. All agent turns are logged to disk with prompt hash, model identifier, token count, and wall-clock latency, providing a full audit trail on the decisions.

A React-based institutional terminal (served from a single static `index.html`) subscribes to a `pod_snapshot.json` file regenerated from the warehouse on each build, and renders the four gates, the BSI timeline, the event-study panels, and the agent log in a Bloomberg-style dark layout. A screenshot of the terminal is omitted from this draft for brevity but is included in the accompanying repository.

12 Risk, Limitations, and Future Work

We enumerate the most material risks to the research thesis and the first-draft implementation.

Data limitations. Most of the first-draft data gap has now closed: as of the April 2026 refresh the warehouse contains 389,099 CFPB complaints across 12 companies (of which 112,405 are BSI-treated BNPL/fintech issuers), 3,666 FinBERT-scored Apple App-Store reviews across the 8 canonical BNPL apps, and five additional macro FRED series (UMCSENT, BAMLH0A0HYM2, STLFSI4, DRSFRMACBS, CSCICP03USM665S). The public-ABS trustee-report ingest completed with 7,998 `abs_tranche_metrics` rows covering Santander Drive (SDART), AmeriCredit (AMCAR), and Exeter (EART) across 2007–2026, of which 2,511 parse cleanly for the 60+-day roll rate and drive the Tier 2 Granger test in §7. Two regex-level coverage gaps remain in the parser and are deferred to v2: (i) the `excess_spread` field returns 0/7,998 hits because the pattern presently requires a trailing `%`-sign that real SDART servicer-exhibit layouts often split across table cells; and (ii) the 60+-day roll-rate anchor is SDART-template-specific, so AMCAR and EART contribute 0/3,776 roll-rate hits despite having otherwise-complete trustee rows. Neither gap affects the paper's headline claim: the Tier 2 falsification test fires on the 2,511 SDART observations alone and still rejects predictive content at $p \geq 0.95$. The Reddit corpus remains pending an API-access approval and is omitted from the refreshed BSI; we treat the Reddit channel as a future additive layer rather than a load-bearing component.

Macro corroboration is structurally unavailable. The original second-draft design-space question was whether a consumer-credit-sensitive macro gate would let the strict three-gate rule fire natively, without the BSI-only super-threshold bypass. §10.3 settles this empirically: five public macro gauges (MOVE, HY OAS, NFCI, STLFSI4, UMCSENT) were simultaneously at or below their long-run medians on the peak-BSI day, and the `gate3_mode=credit` variant (NFCI-gated, with NFCI's full 1971–2026 weekly history) reproduces the zero-approval result. The architectural response — the BSI-only super-threshold bypass at $|z| \geq 10$, documented in §10.2 — has now been

implemented in both the live compliance engine (`agents/compliance_engine.py`) and the backtest driver (`backtest/event_study.py::evaluate_three_gates`). The paper's current finding is therefore no longer counterfactual: the bypass fires natively on the Reg Z window and produces the +5.33 pp TRS-vs-naive spread as an in-architecture result, with an explicit Type-I-premium disclosure attached to every bypassed decision.

AFRMT trustee data is not public (and that is the thesis). The original plan was to re-run Granger causality against the AFRMT 60+-day delinquency roll rate extracted from Affirm Asset Securitization Trust 10-D filings. In implementing that pipeline we discovered that Affirm's securitizations are *144A private placements*: Affirm Holdings (CIK 0001820953) has filed zero 10-D or ABS-15G filings on EDGAR across the entire 2020–2026 sample. The same holds for Afterpay (CIK 0001512673, Block, Inc. parent). The trustee reports for AFRMMT do exist — they are sent directly to the accredited 144A investor base — but are not public and not scrapable. This is *the thesis*: the BNPL-ABS secondary market is definitionally opaque to the public, which is why a BSI-vs-macro-gauge blind spot exists in the first place. We expand the backtest infrastructure to cover public-ABS sponsors (SDART, AMCAR, EART, CACC) as cross-sectional comparators and document the AFRMMT gap as a structural feature of the off-ledger measurement problem this paper addresses, rather than as an implementation gap.

Threshold calibration disclosure. The BSI z-threshold ($+1.5\sigma$), MOVE threshold (120 absolute / P_{85} dynamic), HY-OAS history window (180 day causal z-score), and NFCI threshold (0.0) are all *ex-ante* defaults chosen before running the 5-window sensitivity, not fit to realised firing patterns. We include this disclosure because the paper reports both positive and negative results on the same calibration: BSI fires hard on Reg Z, macro gauges uniformly do not, and the zero-approval count is invariant across threshold specifications. The cross-specification invariance is itself the evidence that the result is robust.

Non-stationarity. Credit-cycle models are famously non-stationary. Our 180-day rolling z-score and 60-day warm-up partly address this by recentreing the distribution, but we do not have a regime-switching or change-point-detection layer in the first-draft pipeline. We would not extrapolate the Granger results naively across regime changes such as a structural shift in the credit-bureau treatment of Pay-in-4.

Scraper fragility. The Reddit and Playwright-based SEC EDGAR pipelines are inherently fragile. A change to the upstream page structure will break parsing silently unless defensive assertions are armed. Our graceful- degradation contract ensures that a broken sub-component does not poison the full BSI, but it does reduce coverage.

Behavioural channel non-stationarity: the AI-search dark-channel shift. The Google-Trends friction bucket is load-bearing for the *direct-admission* leg of the BSI (§6) — queries of the

form *"i can't pay affirm"*, *"affirm hardship program"*, and the eight other first-person distress phrases listed in the Trends taxonomy. This channel presumes that a borrower in distress opens a search engine and types a text query that Google's consumer-scale aggregation service then registers. That presumption is weakening in real time. Industry-survey data from 2026 indicates that roughly 37% of US consumers now begin informational queries inside a conversational AI assistant (ChatGPT, Claude, Gemini, Perplexity) rather than a traditional search engine; the share is higher for younger cohorts, which is precisely the BNPL-heavy demographic. These AI-mediated queries are *not* surfaced in Google Trends or any equivalent public aggregate: they land in private model-provider logs that are not disclosed to third parties, rate-limited, or sold as a commercial dataset. The direct-admission bucket therefore faces a structural "dark-channel" leak: a borrower whose first-person distress query goes to an AI chat rather than to Google is, by construction, invisible to our sensor. This is a harder version of the general-non-stationarity problem: the *channel itself* is migrating, not just the within-channel distribution. Two mitigations are possible without closing the gap fully. First, re-weighting the direct-admission bucket's ω_{DA} coefficient downward as the AI-search share rises, in proportion to a published industry-benchmark series, would keep the BSI honest at the level of *current* coverage rather than *historical* coverage. Second, a secondary signal extracted from public AI-evaluation datasets (where conversational transcripts are released under research licenses — e.g. LMSYS Arena, OpenAssistant) could partially reconstruct the first-person-distress query distribution, at the cost of considerable selection-bias risk. Neither is implemented in the first draft; we flag the shift here because it will be the dominant empirical-validity concern on any v2 with post-2026 data, and any reader extending the BSI to 2027+ should treat the direct-admission bucket as structurally censored rather than simply noisy.

Regulatory arbitrage. The EU CCD II transposition is the single most material regulatory catalyst on our forward calendar. It is also exactly the kind of event that BNPL architects will seek to mitigate via product restructuring (for example, by re-pricing Pay-in-4 products into short-duration revolving lines that fall under different regulatory regimes). A mitigated catalyst is a weaker one.

Strict-threshold Type-II risk. As discussed in §10, the strict three-gate rule fires zero days in our event study. This is a genuine limitation if a single true-positive would have paid for the strategy; in that sense we have an un-observable Type-II error rate. A dynamic-threshold specification of Gate 3, or a partial-credit scoring of the conjunctive rule (e.g. "two of three gates plus a 0.5-weighted fourth") would restore firing density at the cost of a less clean decision boundary. We report this as a design-space tension rather than resolving it in this draft.

No execution integration. The first-draft pod does not include a broker integration. All P&L is simulated from historical public data. The paper makes no claim to a deployable institutional strategy; the contribution is methodological and the implementation is a research prototype.

13 Robustness, Placebo Tests, and Limitations

The preceding empirical sections report positive evidence for the paper's thesis. This section collects the complementary robustness analyses: checks against cost-of-carry degradation, basis risk, a placebo test on non-BNPL macro-stress episodes, sample-size and multiple-comparisons concerns, survivor bias in the consumer-complaint channel, and model risk on the reduced-form hazard specification.

13.1 Cost of carry on the junior-ABS TRS

The Total-Return Swap expression is funded via a bank-counterparty financing leg whose quoted spread over Secured Overnight Financing Rate (SOFR) typically falls in the 75–150 basis-point range for structured-credit receivables of the relevant tranche seniority. Financing spreads at the upper bound of this range compress realised excess return as holding period lengthens, so the institutional expression is designed around a 30–60 trading-day holding window anchored on the regulatory catalyst (§9). Breakeven analysis at the 150 bp upper financing bound indicates that the event-study alpha survives at all five tested windows, with Sharpe ratios compressed by approximately 0.12 relative to the unfunded benchmark. The cost-of-carry risk is therefore first-order quantitative but does not invert the directional result on the windows studied.

13.2 Basis risk and hedge imperfection

The preferred expression pairs long-protection on the AFRMT junior tranche with no equity-side hedge, on the grounds that BNPL-issuer equity is dominated by retail-flow volatility whose covariance with ABS credit performance is small and unstable. Any future extension that adds an equity leg — for example, a long position in American Express as a prime-credit beta neutraliser drawn from Tier 5 of the framework in §5 — introduces basis risk between AXP credit spreads and AFRMT junior loss severity that would become a first-order risk factor. Such an extension would require quarterly re-estimation of the cross-spread hedge ratio and active management of the residual tracking error; the present paper does not pursue this extension and flags it as a future work item.

13.3 Placebo test on the 3-gate system

A placebo test was conducted in which the three-gate decision logic was applied to 40 macro-stress episodes drawn from 2018–2021 — a regime predating the Regulation Z compliance cliff and, for most of the window, predating the commercial emergence of the Pay-in-4 product at scale. The 40 placebo dates consist of sharp equity drawdowns, HY OAS widening episodes, and Treasury-volatility spikes none of which was triggered by a BNPL-specific catalyst. The strict three-gate consensus fired on zero of the 40 placebo dates. The $|\mathrm{BSI}_z| \geq 10$ super-threshold bypass did not arm on any of them: the peak $|\mathrm{BSI}_z|$ realised across the full placebo window was 2.41σ, well below

the 10σ trigger and well below even the unconditional $+1.5\sigma$ gate-one threshold. The null result on non-BNPL macro-stress dates is interpreted as evidence that the system is selecting on BNPL-specific consumer distress rather than reacting to generic macroeconomic risk-off regimes. This is the desired behaviour: a system whose trade thesis is "hidden BNPL-specific leverage" should not fire on a regime in which BNPL volume was structurally too small to carry hidden leverage in the first place.

13.4 Sample-size and multiple-comparisons concerns

The event-study sample comprises five windows and the Granger null ($p \geq 0.95$ at every tested lag 4–8 against the Tier 2 subprime-auto composite, $p > 0.69$ against the Tier 3 HYG proxy) is not a powered test against the strict alternative of non-zero predictive content, as §7.1 makes explicit via the numerical minimum-detectable-effect calculation ($\Delta R^2 \approx 2.6$ to 3.3% at $n = 399$, $\alpha = 0.05$, power 0.80). We therefore do *not* treat the non-rejection as corroboration of orthogonality; it is consistent with zero predictive content and with any incremental explanatory power below the MDE. The paper's illustrative empirical discussion — the +5.33 percentage-point cumulative spread in Table 10 and the 12,838-complaint raw-count pulse of 17 January 2025 (v1 scorer: $+27.4\sigma$) — rests on event-window economics identified on a known catalyst date under a post-hoc threshold (§10.2), not on rejection of any aggregate p-value, and is labelled accordingly throughout. The multiple-comparisons exposure from testing five lags against two target series is acknowledged; a Bonferroni-adjusted rejection threshold would require $p < 0.005$, which is not approached by any tested specification.

13.5 Survivor bias in CFPB complaint data

The Consumer Financial Protection Bureau complaint stream is a self-reported channel: observations are generated by consumers who both possess awareness of the CFPB dispute channel and elect to use it in preference to direct-to-issuer resolution. This population systematically under-represents the most financially distressed cohort, who in the empirical consumer-finance literature typically enter default without filing a regulatory complaint. The BSI is accordingly interpreted throughout this paper as a leading behavioural sensor on the subpopulation that retains the agency to complain, rather than as a comprehensive default forecaster. The distinction is consequential for the interpretation of the 17 January 2025 Regulation Z pulse (12,838 filings against a $\sim$58/day baseline): the event documents an escalation in dispute-channel activity among agency-retaining consumers, not a proportional escalation in realised default rates across the full distressed population.

13.6 Model risk on the reduced-form hazard specification

The Jarrow–Turnbull affine-hazard form

$$\lambda(t) \;=\; \lambda_0 \exp\!\big(\beta_{\mathrm{BSI}}\,\mathrm{BSI}_z(t) + \beta_{\mathrm{MOVE}}\,\mathrm{MOVE}(t)\big)$$

imposes a log-linear assumption on the transmission of stress inputs to the instantaneous default hazard. A fully structural model that represented tranche-level waterfall mechanics, over-collateralisation triggers, and cumulative-loss step-downs endogenously would provide sharper tail-risk estimates at the cost of additional parametric assumptions and additional data requirements — in particular, deal-level waterfall documents that are structurally unavailable for AFRMT by the same 144A disclosure gap discussed in §12. The reduced-form specification is adopted here as the parametrically parsimonious first-pass model that the available data can calibrate. Its use is consistent with the standard practice in the reduced-form credit-pricing literature descending from Jarrow and Turnbull (1995) and Duffie and Singleton (1999).

14 Conclusion

This paper has argued a narrow and specific claim: consumer BNPL is an off-ledger consumer-credit channel whose stress is measurable via a construct-validity-disclosed behavioural sensor built from CFPB complaints and MOVE Treasury-volatility, gated by coverage and residualised against origination-volume mechanics. The paper deliberately does *not* claim that the BSI is a tradable alpha instrument or that the implemented architecture would have prospectively discovered any of the five catalyst events in the sample.

The contributions are methodological. First, a coverage-gated daily composite whose construction is disclosed sufficiently for independent rebuild (§6, repository `github.com/vermasidd1502/bnpl-trap`). Second, a construct- validity decomposition (§3) that names and bounds what a complaint-based sensor can and cannot separate from confounding mechanisms. Third, a falsification gauntlet (§7) consisting of a bivariate Granger F-test with an explicit minimum-detectable-effect calculation (MDE $\Delta R^2 \approx 2.6$ to 3.3% at $n = 399$, $\alpha = 0.05$, power 0.80), three pre-registered placebo sensors, and a local-projection impulse-response specification at 2–8 week horizons. Fourth, a pod-architecture illustration (§10) reporting the single largest raw-count complaint pulse in the sample — 12,838 filings on 17 January 2025 against a trailing $\sim$58/day baseline, a ratio of order 221× — under a post-hoc bypass rule whose provenance we now disclose in §10.2 rather than present as a discovered predictive filter.

We failed to reject the Granger null at every tested lag against both target series. Given the MDE disclosed in §7.1 we do *not* interpret the non-rejection as evidence of orthogonality; the available sample cannot distinguish zero predictive content from incremental ΔR^2 below approximately 3%. An earlier draft framed the null as a falsification headline supporting a "Subprime-2.0 opacity" thesis; we withdraw that framing in v2 and point readers to `docs/v1_retrospective.md` for the

full list of framing-level withdrawals. The same v1 draft reported the MOVE-dominated Granger specification at $p < 10^{-4}$; we now regard that as a MOVE-artefact and have retained the refreshed coverage-gated composite instead.

The most direct extensions of this work are (i) the origination- residual BSI scorer referenced in the roadmap document (`docs/v2_roadmap.md`, Phase C) and re-evaluation of the event-window economics under that scorer; (ii) the AFRMT, SQ, and PYPL originations panels from 10-Q filings needed to calibrate the residual (`docs/v2_roadmap.md`, Phase B); and (iii) the three pre-registered placebo sensors (§7.2).

Two open questions are structural. The first is whether the BSI signal is genuinely BNPL-specific or a general consumer-credit- distress gauge read through a BNPL-complaint channel — a question that cannot be answered from the 2018–2026 CFPB sample alone and that the three placebos attempt to bound. The second is whether an alternative-data behavioural sensor of this family has any role in institutional credit-stress monitoring of off-ledger consumer lending. The claim this paper supports is narrower: *if* such a sensor is going to be built, it should be built and disclosed in the specific way we describe here. The broader policy and measurement agenda — integration with bureau-invisible-credit reporting reform, treatment under the EU Consumer Credit Directive II transposition, and interaction with the regulatory fragility of the CFPB complaint database itself — is left to future work.

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A Reproducibility

The full codebase is a Python package with a single-file `Makefile` that chains all stages:

- `make setup` — creates a `uv` virtual environment and installs the package in editable mode with development extras.
- `make ingest` — runs every data-ingestion module (FRED, SEC EDGAR, CFPB, Reddit, Google Trends, options chain, short interest) into DuckDB.
- `make bsi` — computes the daily BSI and persists it to `bsi_daily`.
- `make validate` — runs Granger causality and rolling out-of-sample tests.
- `make backtest` — runs the event-study simulator and emits per-window P&L CSVs.
- `make pod` — executes one end-to-end pass of the LangGraph pod, logging every agent decision.
- `make snapshot` — rebuilds the React terminal's `pod_snapshot.json`.
- `make paper` — regenerates every figure in this document and compiles the LaTeX.

Each target is idempotent and writes intermediate CSVs / Parquet to disk, so downstream stages can be re-run against cached upstream outputs. A clean clone-and-run on a machine with Python 3.11+, `uv`, and a working TEX distribution should reproduce every number and figure in this paper via:

```
git clone <repo> && cd bnpl-pod && make setup ingest bsi validate backtest paper
```

B Illustrative Equity-Side Pricing Apparatus (Not Gating Reported Results)

This appendix records the two equity-side pricing layers that an earlier draft carried in §8. Neither layer gates any number reported in the main body of this paper — the reported trade expression is a TRS on a junior ABS tranche with no equity hedge — and both are demoted to this appendix so that the main- text pricing discussion reads as a single credit-side argument (Jarrow–Turnbull

reduced-form hazard with affine BSI / MOVE exposures). We retain the specifications here because (i) the Squeeze-Compression Premium SCP_t is reported as non-gating telemetry in every pod decision record (§9), and (ii) an equity-side extension of the overlay would activate this apparatus without further specification work. Readers evaluating the paper's empirical claims should skip this appendix.

B.1 Heston Stochastic-Volatility Layer and the Squeeze-Compression Premium

The equity microstructure of the Architect tier governs the risk of a squeeze in any secondary equity hedge that a future extension may wish to carry. We take Heston (1993) dynamics

$$dS_t = \mu S_t \, dt + \sqrt{v_t} \, S_t \, dW_t^S, \tag{7}$$

$$dv_t = \kappa(\theta - v_t) \, dt + \sigma_v \sqrt{v_t} \, dW_t^v, \tag{8}$$

with $dW^S \, dW^v = \rho \, dt$. We calibrate $(\kappa, \theta, \sigma_v, \rho, v_0)$ daily against the AFRM listed-option surface and compute the Squeeze-Compression Premium

$$\mathrm{SCP}_t \;=\; \mathbb{E}^{\mathbb{Q}}\big[\,(S_{t+21} - K_{0.25})^+\,\big] \;-\; \mathbb{E}^{\mathbb{Q}}\big[\,(K_{0.75} - S_{t+21})^+\,\big], \tag{9}$$

where $K_{0.25}$ and $K_{0.75}$ are the 25% and 75% risk-neutral quantiles of S_{t+21}. SCP_t is the difference between the right-tail call value and the left-tail put value at 21 business days: positive values indicate a right-skewed risk-neutral distribution (squeeze risk elevated), negative values indicate left-skew (crash risk dominant). The z-score of SCP_t is computed and logged in every pod decision record as *non-gating telemetry*; it does not block approval of the TRS expression in §9.

B.2 Squeeze Defense Layer

A complementary Squeeze Defense Layer consumes three microstructure inputs directly: (i) FINRA-reported short interest as a fraction of free float, (ii) borrow-utilisation from public stock-loan aggregators, and (iii) the 25-delta implied-volatility skew of the listed option chain. The layer produces a binary *veto* flag that would suppress any equity-side hedge when at least two of the three signals exceed their regime-specific percentile thresholds. The design defeats the most common failure mode of a credit-informed equity short — the funding squeeze that impaired multiple hedge-fund theses in 2021 — without weakening the credit-side view. The veto is held in reserve for a future equity-side extension of the overlay and is not active in any result reported in the main body of this paper.